

VMAX3 Fundamentals



© Copyright 2015 EMC Corporation. All rights reserved.

Support Contact: [Education Services](#)

Welcome to VMAX3 Fundamentals.

Copyright ©2015 EMC Corporation. All Rights Reserved. Published in the USA. EMC believes the information in this publication is accurate as of its publication date. The information is subject to change without notice.

THE INFORMATION IN THIS PUBLICATION IS PROVIDED "AS IS." EMC CORPORATION MAKES NO REPRESENTATIONS OR WARRANTIES OF ANY KIND WITH RESPECT TO THE INFORMATION IN THIS PUBLICATION, AND SPECIFICALLY DISCLAIMS IMPLIED WARRANTIES OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE.

Use, copying, and distribution of any EMC software described in this publication requires an applicable software license. The trademarks, logos, and service marks (collectively "Trademarks") appearing in this publication are the property of EMC Corporation and other parties. Nothing contained in this publication should be construed as granting any license or right to use any Trademark without the prior written permission of the party that owns the Trademark.

EMC, EMC² AccessAnywhere Access Logix, AdvantEdge, AlphaStor, AppSync ApplicationXtender, ArchiveXtender, Atmos, Authentica, Authentic Problems, Automated Resource Manager, AutoStart, AutoSwap, AVALONidm, Avamar, Bus-Tech, Captiva, Catalog Solution, C-Clip, Celerra, Celerra Replicator, Centera, CenterStage, CentraStar, EMC CertTracker, CIO Connect, ClaimPack, ClaimsEditor, Claralert, cLARIiON, ClientPak, CloudArray, Codebook Correlation Technology, Common Information Model, Compuset, Compute Anywhere, Configuration Intelligence, Configuresoft, Connectrix, Constellation Computing, EMC ControlCenter, CopyCross, CopyPoint, CX, DataBridge, Data Protection Suite, Data Protection Advisor, DBClassify, DD Boost, Dantz, DatabaseXtender, Data Domain, Direct Matrix Architecture, DiskXtender, DiskXtender 2000, DLS ECO, Document Sciences, Documentum, DR Anywhere, ECS, eInput, E-Lab, Elastic Cloud Storage, EmailXaminer, EmailXtender, EMC Centera, EMC ControlCenter, EMC LifeLine, EMCTV, Enginuity, EPFM, eRoom, Event Explorer, FAST, FarPoint, FirstPass, FLARE, FormWare, Geosynchrony, Global File Virtualization, Graphic Visualization, Greenplum, HighRoad, HomeBase, Illuminator, InfoArchive, InfoMover, Infoscapes, Infra, InputAccel, InputAccel Express, Invista, Ionix, ISIS, Kazeon, EMC LifeLine, Mainframe Appliance for Storage, Mainframe Data Library, Max Retriever, MCx, MediaStor, Metro, MetroPoint, MirrorView, Multi-Band Deduplication, Navisphere, Netstorage, NetWorker, nLayers, EMC OnCourse, OnAlert, OpenScale, Petrocloud, PixTools, Powerlink, PowerPath, PowerSnap, ProSphere, ProtectEverywhere, ProtectPoint, EMC Proven, EMC Proven Professional, QuickScan, RAPIDPath, EMC RecoverPoint, Rainfinity, RepliCare, RepliStor, ResourcePak, Retrospect, RSA, the RSA logo, SafeLine, SAN Advisor, SAN Copy, SAN Manager, ScaleIO Smarts, EMC Snap, SnapImage, SnapSure, SnapView, SourceOne, SRDF, EMC Storage Administrator, StorageScope, SupportMate, SymmAPI, SymmEnabler, Symmetrix, Symmetrix DMX, Symmetrix VMAX, TimeFinder, TwinStrata, UltraFlex, UltraPoint, UltraScale, Unisphere, Universal Data Consistency, Vblock, Velocity, Viewlets, ViPR, Virtual Matrix, Virtual Matrix Architecture, Virtual Provisioning, Virtualize Everything, Compromise Nothing, Virtuent, VMAX, VMAXe, VNX, VNXe, Voyence, VPLEX, VSAM-Assist, VSAM I/O PLUS, VSET, VSPEX, Watch4net, WebXtender, xPression, xPresso, Xtrem, XtremCache, XtremSF, XtremSW, XtremIO, YottaYotta, Zero-Friction Enterprise Storage.

Revision Date: December 2015

Revision Number: MR-1WP-VMAXFD.5977.2.0

Course Overview

Description

This course provides an introduction to EMC VMAX3 storage systems. It includes an overview of the VMAX3 architecture, features, and functionality.

Audience

This course is intended for professionals who will be positioning, designing, deploying, managing, and supporting a solution using VMAX3. It is also suitable for anyone seeking to learn the basics of VMAX3.

Objectives

Upon completion of this course, you should be able to:

- Discuss the VMAX3 solution offered by EMC
- Identify key use cases supported by the VMAX3
- Summarize the product architecture
- List hardware and software options
- Describe features and hardware/software modules
- Describe management options and capabilities

This course covers the EMC VMAX3 series. It includes the EMC VMAX3 series models, architecture, features, functions, capabilities, and management.

Module: VMAX3 Introduction

Upon completion of this module, you should be able to:

- Describe the VMAX3 solution
- Describe the benefits of the VMAX3 solution
- Identify key use cases

This module focuses on the VMAX3 product benefits and advantages, and how they solve the needs of customers. An overview of the VMAX3 family will be presented.

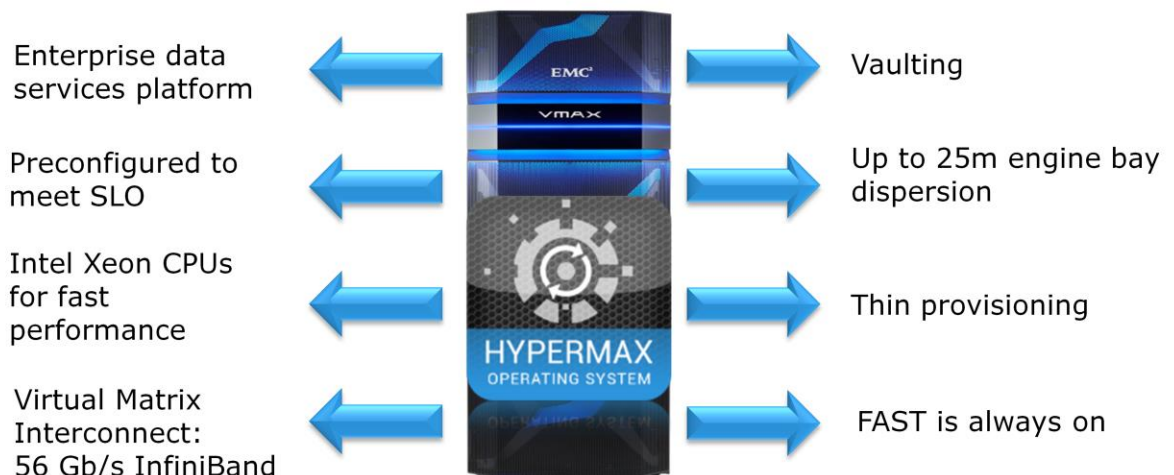
Lesson: VMAX3 Solution

This lesson covers the following topics:

- What is VMAX3?
- Business and technical challenges
- VMAX3 benefits

This lesson covers an overview of the VMAX3 solution and its benefits.

What is VMAX3?



© Copyright 2015 EMC Corporation. All rights reserved.

VMAX3 Fundamentals

EMC²

5

VMAX3 is a data services platform that specifically addresses the new requirements of the modern data center. Some of its important features are:

- VMAX3 is an enterprise data services platform built for the hybrid cloud.
- VMAX3's HYPERMAX OS is the industry's first storage and hypervisor converged operating system. It combines industry-leading high availability, I/O management, quality of service, data integrity validation, storage tiering and data security with an open application platform.
- VMAX3 is preconfigured at the factory to meet predetermined Service Level Objectives.
- Intel Xeon CPUs are utilized for providing fast processing of I/O at all (Front-End, Back-End, Cache) points in the array.
- VMAX3 has an InfiniBand Virtual Matrix Interconnect which supports up to 56 Gb/s speed.
- Vaulting is done to Flash I/O modules thereby reducing battery requirements, weight, and space.
- Engine bays can be separated by up to 25 meters to fit into today's data centers.
- VMAX3 utilizes standard 24" wide racking. Third party racking is now standard (for approved racks) for all systems.
- It has a high performing native 6Gb/s SAS back-end infrastructure.
- All provisioning is now thin, providing the best space utilization and flexibility.
- Fully Automated Storage Tiering (FAST) constantly monitors and moves data to the appropriate storage tier based on access and requirements.

Business and Technical Challenges



© Copyright 2015 EMC Corporation. All rights reserved.

VMAX3 Fundamentals

EMC²

6

The key challenges that today's businesses face are:

- Handling the enormous data growth
- Consolidation and virtualization of data
- Providing 24/7 operation
- Security and compliance

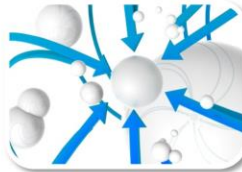
Business Needs and VMAX3 Core Business Benefits

24/7
365

Non-disruptive
operation



Virtualization



Single-action
provisioning



Scalability



Capacity
management



Ease of
management



Lower TCO

EMC²

VMAX3 Fundamentals

7

VMAX3 is the world's most trusted storage platform. It meets all the storage requirements for today's business needs. Key reasons why customers choose the VMAX3 family are:

- VMAX3 delivers 'always-on' availability for non-disruptive operation of businesses and improves service delivery.
- The VMAX3 product family is purpose-built for virtual environments. Integration with virtualized servers enable virtual environments.
- The high-end capabilities are combined with unmatched levels of scale, ease of use, and automation.
- It offers Fully Automated Storage Tiering (FAST), which automatically optimizes performance to meet service level requirements.
- Users can also start small and scale-out resources to provide performance on demand, with predictable service levels.
- VMAX3 helps to achieve space efficiency with virtual provisioning. This enables the customers to easily grow and reclaim storage.
- VMAX3 enables businesses to get rid of storage management at a granular level and also dedicating man power for the same.
- It also enables businesses to deliver higher service levels through scale-out and tiering at the lowest Total Cost of Ownership (TCO).

VMAX3 Virtualization Business Benefits



© Copyright 2015 EMC Corporation. All rights reserved.

VMAX3 Fundamentals

8

Virtualization is one of the important benefits of VMAX3. Symmetrix and VMAX3 have a history of supporting mission-critical applications in Tier 1 environments across mainframes, open systems, and files, when deployed with eNAS. As companies now move to the cloud, VMAX3 with HYPERMAX OS 5977 enables businesses to virtualize Tier 1 applications with confidence.

With VMAX3, EMC is bringing these same capabilities to newer cloud-based environments with support for virtual machines, virtual desktops, and object data. As customers look to virtualize Tier 1 applications, doing so with VMAX3 using EMC's proven legacy, simplifies the process.

Lesson Summary

During this lesson the following topics were covered:

- What is VMAX3?
- Business and technical challenges
- VMAX3 benefits

This lesson covered VMAX3 solutions and its benefits.

Lesson: Use Cases for VMAX3 Solution

This lesson covers the following topics:

- Key application use cases
- Cross-product solution use cases

This lesson covers use cases for VMAX3 solutions.

VMAX3 Microsoft Use Cases: Exchange and SQL



VMAX3 benefits:

- TimeFinder SnapVX snapshots help in efficient use of storage resources and optimize protection
- VMAX3 performance capability is optimal for heavy Exchange workloads



VMAX3 benefits:

- FAST – Fully Automated Storage Teiring improves performance and resource utilization while lowering the overall costs

EMC²

© Copyright 2015 EMC Corporation. All rights reserved.

VMAX3 Fundamentals

11

The Key Application Use Cases of VMAX3 are: Microsoft Exchange Use Case, Microsoft SQL Server Use Case, Oracle Use Case and SAP Use Case

Microsoft Exchange:

The pressure in **Exchange** environments to eliminate backup windows and reduce recovery times is stronger than ever before. This problem is easily overcome by using TimeFinder SnapVX advanced technologies such as snapshots that offers scalability and ease of use features, and the snapshots are consistent by default.

It also helps in efficient use of storage resources and optimize protection for Microsoft Exchange. EMC TimeFinder SnapVX software delivers instant and storage-consistent point-in-time replicas of host devices that can be used for backup and recovery. The VMAX3 storage array's performance is optimal for any heavy exchange workload.

Microsoft SQL Server:

Traditionally, the best practices for optimizing storage performance involved manual, tedious and resource-intensive processes. VMAX3 FAST allows **SQL** administrators to leverage a hands-off mechanism for optimizing the performance of the most demanding applications.

Automating the movement of data between storage tiers saves both time and resources. VMAX3 FAST eliminates the need to spend countless hours manually monitoring and analyzing the data to determine a storage strategy and then maintaining, relocating and migrating LUNs to the appropriate storage tiers.

VMAX3 Use Cases: Oracle and SAP Applications



ORACLE®

VMAX3 benefits:

- Service Level Objective based storage provisioning
- Leverage Flash drives and automatic storage tiering to reduce number of spindles required



SAP

VMAX3 benefits:

- Faster with less latency and a smaller footprint
- Simplify storage with choice in Service Level Offerings

EMC²

© Copyright 2015 EMC Corporation. All rights reserved.

VMAX3 Fundamentals

12

Listed are some of the benefits offered by VMAX3 for **Oracle** applications:

- **Online Transaction Processing (OLTP)** database applications tend to be mission-critical and usually have stringent I/O latency requirements.
- Traditionally, these OLTP databases are deployed on a huge number of rotating spindles to meet the low I/O latency requirement. Consequently, the effective capacity utilization of these spindles is very low. VMAX3 and Flash drives reduces the need to buy more drives to keep up with database growth.
- Also with VMAX3, **FAST** automatically and non-disruptively migrates hot and cold data between the available storage tiers, thereby improving the effective storage utilization. VMAX3 simplifies and streamlines storage provisioning for Oracle with **SLO** and also provides a Performance Analytics Tool for the Oracle Database Administrator.

Now, let us see some of the benefits offered by VMAX3 for **SAP** applications:

- The common business requirement in SAP environments is reducing TCO while improving performance and service level delivery. Frequently, responsiveness to sensitive SAP applications has deteriorated over time due to increased data volumes, unbalanced data stores, and changing business requirements.
- By using VMAX3, SAP deployments can gain a significant performance boost without the need to redesign the applications, adjust the data layouts, or reload significant amounts of data. With automated sub-LUN level tiering, FAST automatically balances the data distribution across the tiers based on service levels that allows capacity and performance optimization.
- **TimeFinder SnapVX** is used to reduce risks from logical errors and human intervention without affecting the SAP system performance.

Cross-product Solution Use Cases

- VMAX3 ProtectPoint use case
- VMAX3 use case: FAST.X extends data services



© Copyright 2015 EMC Corporation. All rights reserved.

VMAX3 Fundamentals

EMC²

13

The Cross-product solution use cases for VMAX3 are:

- ProtectPoint use case
- FAST.X – extends data services use case

Cross-product Solution Use Case: ProtectPoint



VMAX3 ProtectPoint

- Direct backup from primary storage to protection storage
- Faster backup and recovery
- Instant access to protected data

Eliminates
application/database
impact

Eliminates separate
backup infrastructure

Reduction in number
of copies

EMC²

© Copyright 2015 EMC Corporation. All rights reserved.

VMAX3 Fundamentals

14

EMC ProtectPoint enables direct backup from primary storage to protection storage. It ensures faster backup and reliable recovery as well as instant access to protected data.

It is designed to address the gap in protection solutions when it comes to mission critical applications that traditional backup and snapshots alone cannot meet including:

- Eliminating the application/database impact
- Eliminating the need for a separate backup infrastructure and giving control back to the database team
- Large scale reduction in number of copies

Cross-product Solution Use Case: FAST.X



FAST.X extends data services

VMAX3 FAST.X

- Extends data services beyond the array and across the data center
- Seamless integration of VMAX storage and heterogeneous arrays
- Extends storage tiering and simplifies SLO management
- Ensures workload optimization without complicated management

EMC²

VMAX3 Fundamentals

15

One of the common business requirement is to provide data services and simplify SLO management across the data center. This is solved using VMAX3's FAST.X by:

- VMAX's data services extend beyond the array and across the data center through FAST.X.
- FAST.X introduces the seamless integration of VMAX storage and heterogeneous arrays.
- FAST.X evolves and extends service level management to other storage platforms and the cloud.
- It extends storage tiering and service level management to other storage platforms (including XTREMIO and 3rd party arrays) and to the cloud (CloudArray), enabling the use of LUNs on external storage as raw capacity.
- It enables SLO management across external arrays enabling easy integration for different appliances as needed by different workloads and requirements. Whether it's integrating XTREMIO for better space efficiency or using CloudArray to off load data to an object store, FAST.X ensures workload optimization without complicated management.

Lesson Summary

During this lesson, the following topics were covered:

- Key application use cases
- Cross-product solution use cases

This lesson covered key application use cases and cross-product solution use cases.

Module Summary

Key points covered in this module:

- VMAX3 solution
- Benefits of the VMAX3 solution
- Key use cases of VMAX3

This module covered VMAX3 solution, its benefits, and key use cases.

Module: VMAX3 Architecture

Upon completion of this module, you should be able to:

- Describe VMAX3 architecture and key terminology
- Identify VMAX3 components
- Describe and compare VMAX3 configurations and models

This module focuses on VMAX3 architecture, terminology, components, configurations, and models.

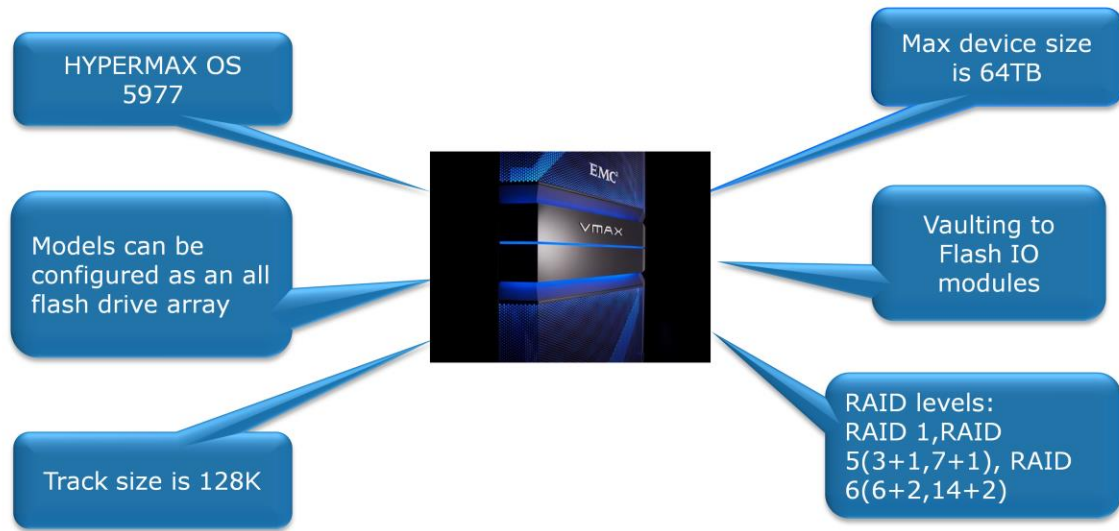
Lesson: VMAX3 Architecture and Components

This lesson covers the following topics:

- VMAX3 architecture
- VMAX3 components

This lesson covers VMAX3 architecture and its components.

VMAX3 Architecture



© Copyright 2015 EMC Corporation. All rights reserved.

VMAX3 Fundamentals

EMC²

20

The VMAX3 arrays introduce the industry's first open storage and hypervisor converged operating system, HYPERMAX OS 5977. The HYPERMAX OS combines industry-leading high availability, I/O management, quality of service, data integrity validation, storage tiering, and data security with an open application platform.

The VMAX3 models can be configured as an all flash drive array. The supported track size is 128K and the allocation of tracks occur only when a write occurs. The maximum host addressable devices or TDEVs are 64K. The maximum device size is 64 terabytes.

With the VMAX3 series, all vaulting is done to Flash IO modules. Heavy lead batteries are replaced by Lithium-Ion batteries for backup. This results in the reduction of space and weight.

RAID protection options are configured at the volume level. The same VMAX3 can employ a variety of protection schemes.

Some of the RAID protection options are:

- RAID 1 withstands failure of one drive within one of the mirrored pairs. The number of required drives is twice the amount required to store data.
- RAID 5 protection is also available. Data blocks are striped horizontally across the members of a RAID 5 group, and each member owns some data tracks and some parity tracks. RAID 5(3+1) and RAID 5(7+1) are supported.
- RAID 6 protects data with failures of up to 2 drives per RAID group. RAID 6(6+2) and RAID 6(14+2) is supported. RAID 6 is ideal for large 7K rpm drives.

VMAX3 Engine

- 4U enclosure
- Two director boards
- Each director board has its own redundant power and cooling
- Two management modules
- 11 IO slots per director



© Copyright 2015 EMC Corporation. All rights reserved.

VMAX3 Fundamentals

EMC²

21

The engine is a critical building block of a VMAX3 system and has a 4U enclosure.

The main components of an engine are: director boards, management modules, fan, power supply, and IO slots. VMAX3 engine consists of two redundant director boards which house global memory, front-end connectivity, back-end connectivity and internal network communication components.

It has two types of management modules. They are – Management Module Control Station (MMCS) and the standard Management Module (MM). The cooling fans are replaceable and use adaptive cooling based on the system activity. The independent director power supplies provide availability. There are 11 IO slots per director.

Drive Enclosures/Drives

- DAE60 3.5" x 60 drives
- DAE120 2.5" x 120 drives
- Allows DAE mixing of types within array

Drive Type	Capacities	Carrier	
		2.5"	3.5"
SSD	200, 400, 800GB, and 1.6TB	Y	Y
15K RPM	300GB	Y	Y
10K RPM	300, 600, and 1200GB	Y	Y
7.2K RPM	2 and 4TB	N	Y

EMC²

© Copyright 2015 EMC Corporation. All rights reserved.

VMAX3 Fundamentals

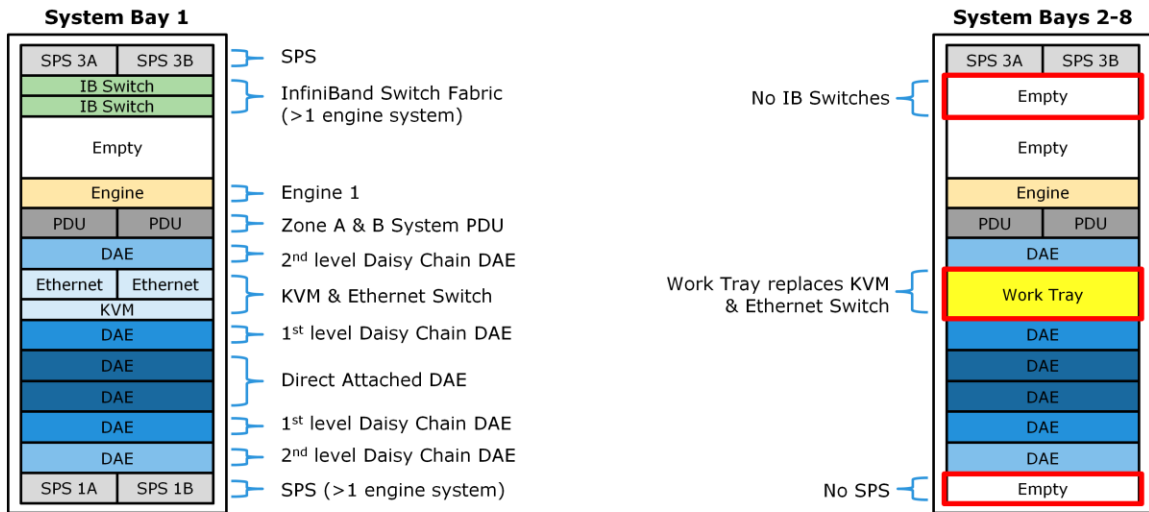
22

DAE stands for disk-array enclosure. It is a rack / shelf that hold a large group of hard disk drives used in the array. They enable the expansion of each array's drive count to provide the storage needed for the expanding needs.

All drives in VMAX3 DAE are 2.5" native, in either size carriers except 7200 RPM drives which are available only in 3.5" carriers. The drives support all flash configurations. These DAEs allow greater number of drives to be added to a rack unit, therefore better utilizing floor footprint.

EMC is continuously evaluating, testing, and qualifying new drives. Check the support matrix for the most current information.

Bay Components – Single Engine Bay



© Copyright 2015 EMC Corporation. All rights reserved.

VMAX3 Hardware Fundamentals

EMC²

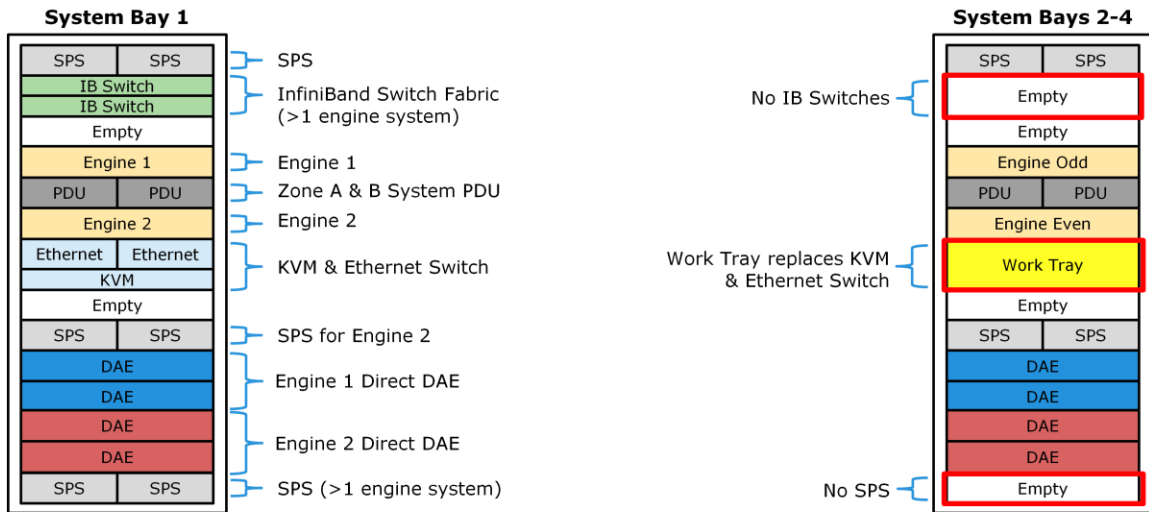
23

The components in System Bay 1 of a single engine bay include:

- Standby Power Supplies (SPSs) to support the engine
- A Fabric and additional SPSs (in multi-engine systems only)
- An engine
- A Power Distribution Unit (PDU)
- A KVM (Keyboard, Video, Mouse)
- A set of Ethernet switches (seen from the rear of the bay)
- Disk Array Enclosures (DAEs) associated with the engine

System Bays 2 through 8 do not have the Fabric and additional SPSs, KVM, or Ethernet switches. A Work Tray is included in System Bays 2 through 8 in place of the KVM.

Bay Components – Dual Engine Bay



© Copyright 2015 EMC Corporation. All rights reserved.

VMAX3 Hardware Fundamentals

EMC²

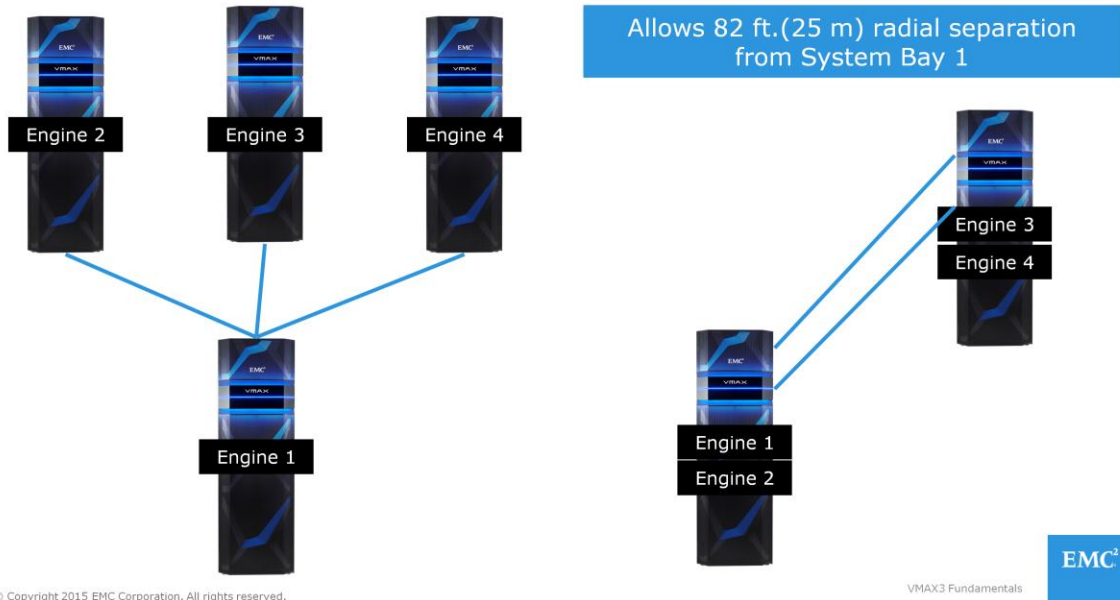
24

The components in System Bay 1 of a dual engine bay include:

- Standby Power Supplies (SPSs) to support the engines
- A Fabric and SPSs (in multi-engine systems only)
- One or two engines
- A Power Distribution Unit (PDU)
- A Keyboard/Video/Mouse (KVM)
- An Ethernet switch
- Disk Array Enclosures (DAEs)

System Bays 2 through 4 do not have the Fabric and its SPSs, KVM, or Ethernet switches. A Work Tray is included in System Bays 2 through 4 in place of the KVM.

Dispersed Arrays



System Bay dispersion allows the separation of any individual or contiguous group of System Bays from System Bay 1. This provides unsurpassed datacenter flexibility in solving floor loading constraints or working around obstacles that might prevent fully contiguous configurations. Dispersion of System Bays 2 through 8 (in single engine racks) and System Bays 2 through 4 (in dual engine racks) is supported in the VMAX3 family of arrays. Longer Matrix Interface Board Enclosure (MIBE) and Ethernet cable bundles allow dispersion up to 82 feet (25 meters). Those adjacent to Engine 1 use copper interconnect and save cost with no loss of functionality.

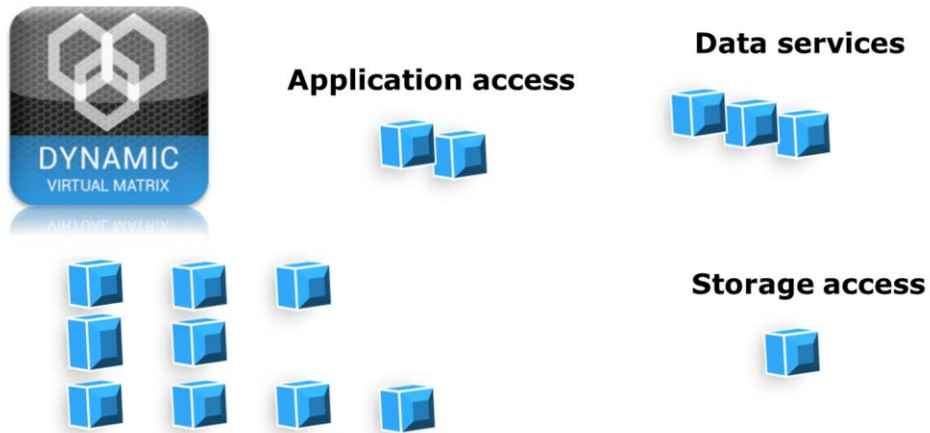
The slide shows examples of array dispersion.

Left: A four engine system – single engine configuration

Right: A four engine system – dual engine configuration

The blue lines on the slide represent multiple InfiniBand and Ethernet links.

Dynamic Virtual Matrix



© Copyright 2015 EMC Corporation. All rights reserved.

VMAX3 Fundamentals



26

The new Dynamic Virtual Matrix feature of VMAX3 allows scaling without compromise. The Dynamic Virtual Matrix is architected for performance at high scale, taking multi-core optimization to a whole new level. Several cores and terabytes of cache are pooled and allocated on demand. Reallocation of cores and cache space is easy and is based on the work load on each site.

HYPERMAX Operating System

Management



Block



File



Cloud Access



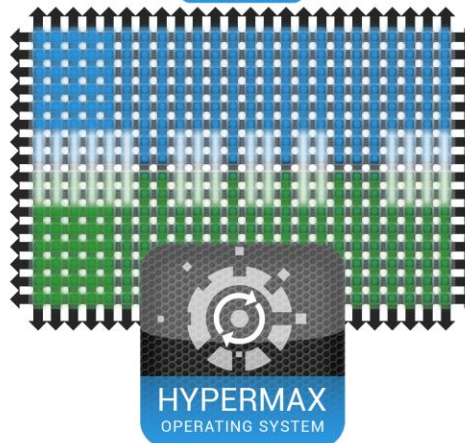
SDDC



Backup



Analytics



EMC²

© Copyright 2015 EMC Corporation. All rights reserved.

VMAX3 Fundamentals

27

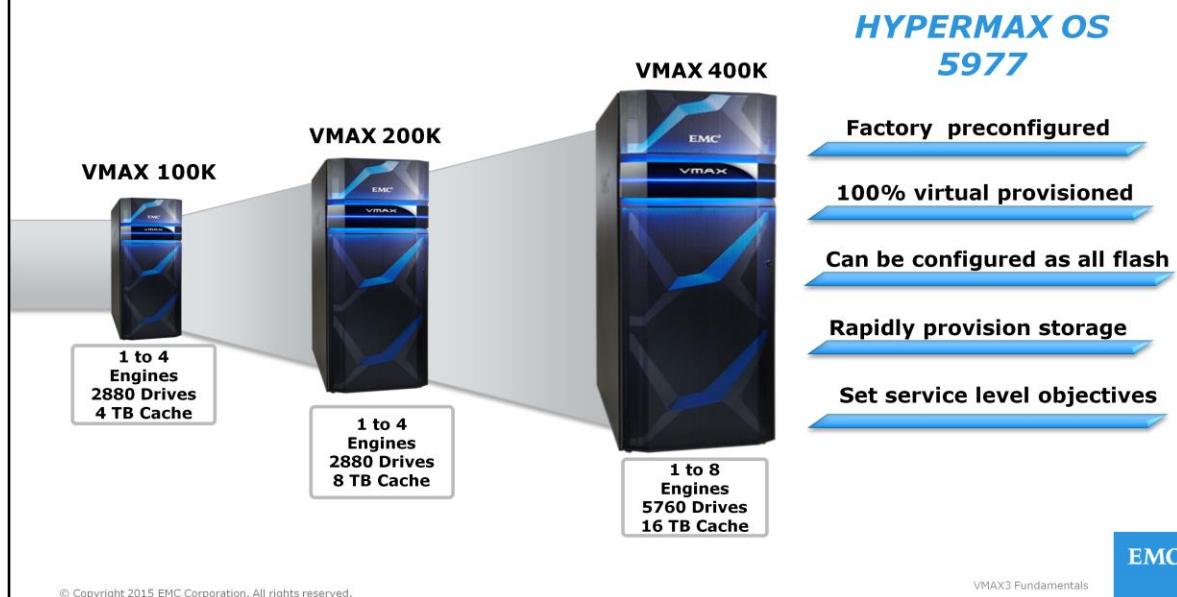
EMC introduces the industry's first storage and hypervisor converged operating system called HYPERMAX OS. It combines industry-leading high availability, I/O management, quality of service, data integrity validation, storage tiering and data security with an open application platform.

HYPERMAX OS features the first real-time, non-disruptive storage hypervisor that manages and protects embedded services by extending the high availability feature of VMAX3 to these services, that traditionally would have run external to the array. It also provides direct access to hardware resources to maximize performance. The hypervisor can be upgraded without disruption.

The embedded storage hypervisor reduces external hardware and networking requirements, delivers higher levels of availability and dramatically lowers latency. The HYPERMAX OS runs on the Dynamic Virtual Matrix, leveraging its scale-out flexibility of cores, cache, and host interfaces.

EMC offers many embedded services including embedded NAS to deliver unified storage on VMAX, as well as embedded FAST functionality and operating environment tools. Strategically, EMC's approach is to open up this capability to the industry, with an initial focus on applications that can leverage the best in class data services of VMAX such as Big Data ETL & Analytics, Cloud gateways, etc.

VMAX3 Family

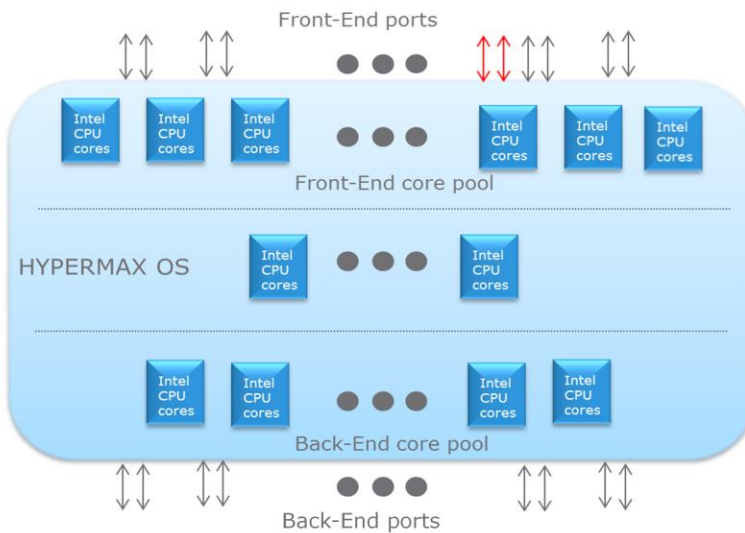


The new VMAX3 family with HYPERMAX OS 5977 consists of three arrays: VMAX 100K, VMAX 200K, and VMAX 400K. All three models can be configured as all flash arrays. The previous versions of the VMAX3 family used an operating system called Enginuity. Starting with VMAX3, the array operating system is called HYPERMAX OS. The HYPERMAX OS has a storage hypervisor that enables embedded data services and applications.

The VMAX3 family arrays are 100% virtual provisioned and pre-configured in the factory. The arrays are built for simple management, extreme performance, and massive scalability with a small footprint. With the VMAX3 family of arrays, storage can be rapidly provisioned with a desired Service Level Objective (SLO).

Since the VMAX3 is preconfigured at the factory, the time from delivery to first I/O is significantly shortened.

VMAX3 Multi-core Emulation



© Copyright 2015 EMC Corporation. All rights reserved.

VMAX3 Fundamentals

29

Multi-core emulation provides additional CPU and physical port utilization capabilities to HYPERMAX OS emulations, extending the existing core architecture and improving the overall performance. One key feature of Multi-core emulation is pre-defined core mappings that allow specification of performance characteristics based on expected I/O profiles and usage. VMAX3 arrays can be configured to be front-end centric (allocating more CPU cores to handle host I/O), back-end centric (allocating more CPU cores to handle disk I/O), or the default baseline configuration where CPU cores are evenly distributed between front and back end.

The slide shows the default Multi-core emulation in VMAX3 arrays. Cores are provided for front end, back end, and for HYPERMAX OS functions. All of the CPU cores on the director work on I/O from all of the ports. This helps ensure VMAX3 director's ports are always balanced.

Lesson Summary

During this lesson the following topics were covered:

- VMAX3 architecture
- VMAX3 components

This lesson covered VMAX3 architecture and its components.

Lesson: VMAX3 Models and Options

This lesson covers the following topics:

- VMAX 100K
- VMAX 200K
- VMAX 400K
- VMAX3 models comparison
- Common features

This lesson covers VMAX 100K, 200K, and 400K arrays. Their features and components have been listed in the table for comparison of the models. The common features of the models have also been listed.

VMAX3 Family: VMAX 100K Architecture

- One to four engines:
 - 2.1 GHz, 24 Intel Xeon cores
 - 512GB, 1TB memory
- 2880 2.5" or 1440 3.5" drives
- 4TB cache
- 1.1PB usable capacity and 128 front-end ports
- Dual Management Module Control Station (MMCS):
 - Integrated service processor
- InfiniBand Fabric:
 - 56 Gb/s speeds and 12 port InfiniBand switches



EMC²

© Copyright 2015 EMC Corporation. All rights reserved.

VMAX3 Fundamentals

32

The EMC VMAX3 family has three models – VMAX 100K, 200K, and 400K.

The VMAX 100K is configured with one to four engines. With the maximum four-engine configuration, the VMAX 100K supports up to 2880 2.5" drives, or up to 1440 3.5" drives, providing up to 1.1 Petabytes of usable capacity. When fully configured, the VMAX 100K provides up to 128 front-end ports for host connectivity. The Internal Fabric Interconnect uses InfiniBand Dual Redundant Fabric with the speed of 56Gbps per port for redundancy and availability. The Infiniband switch in the VMAX 100K has 12 ports. The VMAX 100K supports up to 4TB Cache.

VMAX3 Family: VMAX 200K Architecture

- One to four engines
 - 2.6 GHz, 32 Intel Xeon cores
 - 512GB, 1TB, 2TB memory
- 2880 2.5" or 1440 3.5" drives
- 8 TB cache
- 2.3PB usable capacity and 128 front-end ports
- Dual Management Module Control Station (MMCS):
 - Integrated service processor
- InfiniBand Fabric:
 - 56 Gb/s speeds and 12 port InfiniBand switches



EMC²

© Copyright 2015 EMC Corporation. All rights reserved.

VMAX3 Fundamentals

33

The VMAX 200K is configured with one to four engines. With the maximum four-engine configuration, the VMAX 200K supports up to (2880) 2.5" drives, or up to (1440) 3.5" drives, providing up to 2.3 Petabytes of usable capacity. When fully configured, the 200K provides up to 128 front-end ports for host connectivity. The Internal Fabric Interconnect uses InfiniBand Dual Redundant Fabric with the speed of 56Gbps per port for redundancy and availability. The Infiniband switch in the 200K has 12 ports. The VMAX 200K supports up to 8TB Cache.

VMAX3 Family: VMAX 400K Architecture

- One to eight engines
 - 2.7 GHz, 48 Intel Xeon cores
 - 512GB, 1TB, 2TB memory
- 5760 2.5" or 2880 3.5" drives
- 16 TB cache
- 4.3PB usable capacity and 256 front-end ports
- Dual Management Module Control Station (MMCS):
 - Integrated service processor
- InfiniBand Fabric:
 - 56 Gb/s speeds and 18 port InfiniBand switches



EMC²

© Copyright 2015 EMC Corporation. All rights reserved.

VMAX3 Fundamentals

34

The VMAX 400K is configured with one to eight engines. With the maximum eight-engine configuration, the VMAX 400K supports up to (5760) 2.5" drives, or up to (2880) 3.5" drives, providing up to 4.3 Petabytes of usable capacity. When fully configured, the 400K provides up to 256 front-end ports for host connectivity. The Internal Fabric Interconnect uses InfiniBand Dual Redundant Fabric with the speed of 56Gbps per port for redundancy and availability. The InfiniBand switch in the 400K has 18 ports. The VMAX 400K supports up to 16TB Cache.

All VMAX3 arrays contain two Management Module Control Systems (MMCS) in System Bay 1. This helps to increase system availability as there are multiple access points to the system for remote access. If there is a failure in either MMCS, the system is able to dial home from the remaining MMCS for remote recovery or diagnose if hardware replacement is required.

VMAX 100K, 200K, and 400K Comparison

Components/Features	VMAX 100K	VMAX 200K	VMAX 400K
No. of engines supported	1 to 4	1 to 4	1 to 8
CPU	Intel Xeon E5-2620-V2, 2.1GHz, 6 core	Intel Xeon E5-2650-V2, 2.6GHz, 8 core	Intel Xeon E5-2697-V2, 2.7GHz, 12 core
No. of CPU cores per engine	24	32	48
Cache system (maximum)	4TB	8TB	16TB
2.5" DAE Configurations (no. of drives)	2880	2880	5760
3.5" DAE Configurations (no. of drives)	1440	1440	2880
Front-End ports	128	128	256
Usable capacity	1.1 P Bu	2.3 P Bu	4.3 P Bu
DVM Interconnect	InfiniBand Dual Redundant Fabric- 56Gbps per port	InfiniBand Dual Redundant Fabric- 56Gbps per port	InfiniBand Dual Redundant Fabric- 56Gbps per port

EMC²

© Copyright 2015 EMC Corporation. All rights reserved.

VMAX3 Fundamentals

35

This table gives a comparison of the three VMAX3 models on various parameters such as number of engines, cache, DAE configurations, etc. All three models can be configured as all-Flash arrays or hybrid arrays, and all share the same software and hardware features. The key differentiator between models is the number of CPU cores per engine and how big they can scale.

The Dual MMCS replaces the Service Processor that was present in earlier VMAX models. The DVM Interconnect in all the models is the InfiniBand Dual Redundant Fabric - 56Gbps per port.

Common Features

Feature	Description
Maximum Drives per Engine	• 720 (2.5") • 360 (3.5")
Drive Options	• Hybrid (mixed drive types) • All-Flash
DAE Mixing	• 60-drive and 120-drive behind an engine • Single increments
Power Options	• Three-Phase Delta (50 amp), WYE (32 amp) • Single-Phase (32 amp)
Dispersion	System Bays 2-8, up to 25 meters from System Bay 1
Vault	Vault to Flash in engine
Racking Options	• Single engine; Dual engine • Third-party racking
Service Access	• Integrated service processor in System Bay 1; Management Module Control Station (MMCS) • Service tray on additional System Bays

© Copyright 2015 EMC Corporation. All rights reserved.

VMAX3 Fundamentals

EMC²

36

Common features throughout the VMAX3 family include– maximum drives per engine; both hybrid and all-Flash, DAE mixing behind engines in single increments, power configuration options, System Bay dispersion, multiple racking options and service access points. Also, Vault to Flash in engine is implemented on the VMAX3 family, which is a change from the previous vaulting process. Service access is provided by a Management Module Control Station (MMCS), which is the integrated service processor located in System Bay 1.

This table shows the common features between the VMAX 100K, 200K, and 400K.

Lesson Summary

During this lesson, the following topics were covered:

- VMAX 100K
- VMAX 200K
- VMAX 400K
- VMAX3 models comparison
- Common features

This lesson covered VMAX 100K, 200K, and 400K arrays. The features and components of the models were compared and the common features were listed.

Module Summary

Key points covered in this module:

- VMAX3 architecture and key terminology
- VMAX3 components
- VMAX3 configurations and models

This module covered VMAX3 architecture, terminology, components, configurations, and models.

Module: VMAX3 Features and Capabilities

Upon completion of this module, you should be able to:

- Describe VMAX3 key features and capabilities
- Identify the usage of key features and capabilities in an IT environment

This module focuses on VMAX3 functions and features such as read/write operations, data integrity, storage efficiency, local & remote replication, integration of eNAS with VMAX3, and security. It also covers the usage of key features and capabilities in an IT environment.

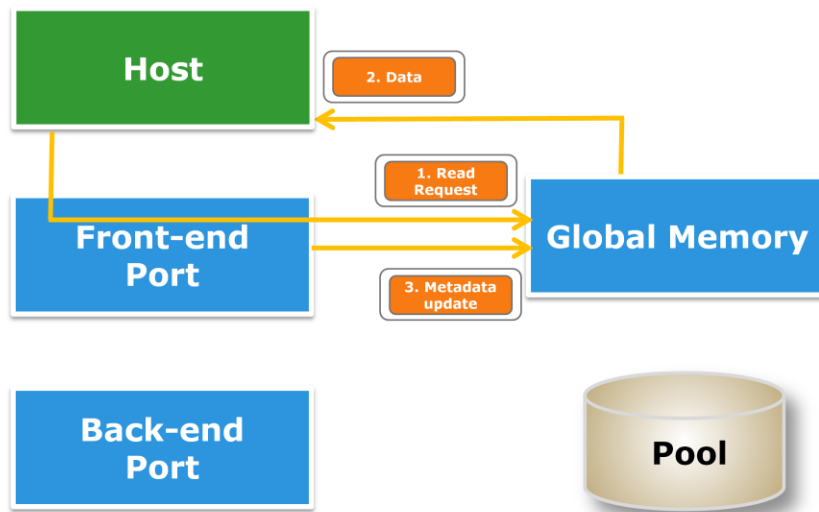
Lesson: Read/Write Operations

This lesson covers the following topics:

- Read operations: Read hit
- Read operations: Read miss
- FlashBoost
- Write operations: Fast Write
- Write operations: Delayed Fast Write

This lesson covers the read operations – read hit and read miss, and FlashBoost. It also covers the write operations – Fast Write and Delayed Fast Write.

Read Operations – Read Hit



© Copyright 2015 EMC Corporation. All rights reserved.

VMAX3 Fundamentals

EMC²

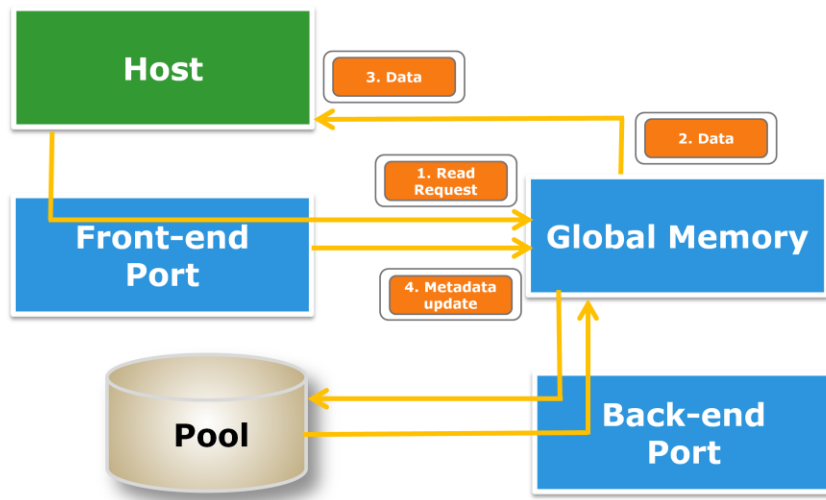
41

In a **read hit** operation, the requested data is directly read from the global memory. The sequences of steps that occur during the read hit operation are:

1. The host issues a read request.
2. The requested data is found in the global memory and is transferred to the host through the VMAX3 front-end port.
3. The metadata is updated in the global memory.

Since the data is available in global memory, there are no delays due to disk seek time and latency.

Read Operations – Read Miss



© Copyright 2015 EMC Corporation. All rights reserved.

VMAX3 Fundamentals

EMC²

42

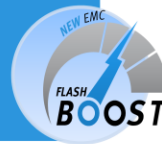
The **read miss** operation is triggered when the requested data is not found in global memory. In this case, the VMAX3 system must search for data from a disk drive in the pool and then transfer it to global memory. The sequences of steps that occur during the read miss operation are:

1. The host issues a read request.
2. The disk is searched for the data and then it is sent to the global memory. While space is created in global memory, the VMAX3 back-end port reads the data from the pool.
3. The Front-End port reconnects with the host and transfers the data to it.
4. The metadata is updated in global memory.

FlashBoost

Services VMAX3 front-end requests directly from back-end drives by bypassing the global cache to service the read miss

Improved efficiency



EMC²

© Copyright 2015 EMC Corporation. All rights reserved.

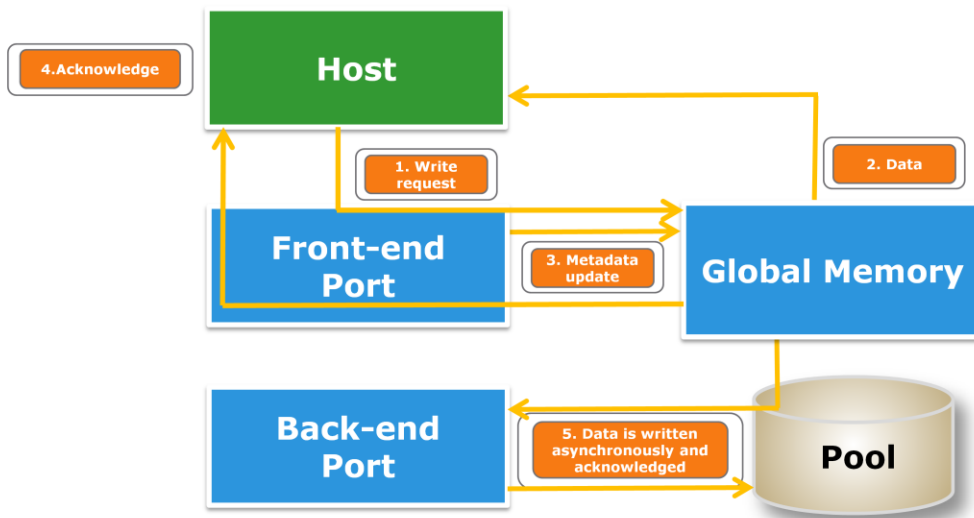
VMAX3 Fundamentals

43

FlashBoost delivers performance acceleration for VMAX3 customers running high demand read-intensive workloads. It maximizes the efficiency of HYPERMAX OS by servicing VMAX3 front-end requests directly from back-end drives. This approach eliminates steps required for processing IO through global cache and reduces the latency for reads, particularly for flash drives.

Performance results vary depending on customer workload. Customers with 100% read miss workloads residing on flash can see up to 100% greater IOPS performance. However, mixed workloads with heavy writes may not see significant performance gains with FlashBoost enabled.

Write Operations – Fast Write



© Copyright 2015 EMC Corporation. All rights reserved.

VMAX3 Fundamentals

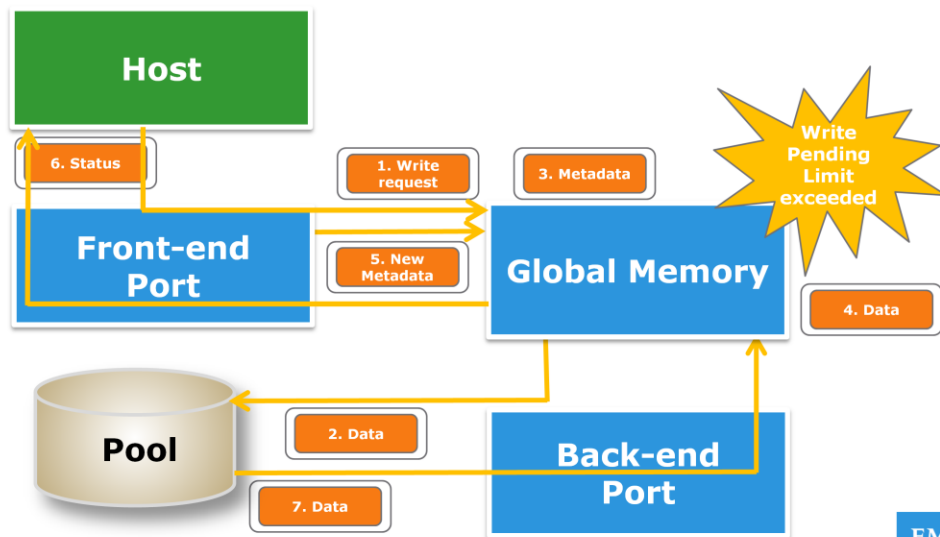
EMC²

44

A **fast write** occurs when there is sufficient global memory space available for the incoming write request. The sequences of steps that occur during a Fast Write operation are:

1. The host issues a write request.
2. The VMAX3 front-end port places the incoming data directly into global memory.
3. The metadata is updated.
4. An acknowledgment is sent to the host indicating that the data has been written.
5. The VMAX3 back-end ports asynchronously de-stage the data from global memory to the pool and send the acknowledgement to the global memory.

Write Operations – Delayed Fast Write



© Copyright 2015 EMC Corporation. All rights reserved.

VMAX3 Fundamentals

EMC²

45

A **delayed fast write** occurs only when the Fast Write threshold has been exceeded. The write threshold, also called the **Write Pending Limit** is the percentage of global memory that contains modified data. The sequences of steps that occur during a Delayed Fast Write operation are:

1. The host issues a write request.
2. The VMAX 3 system de-stages some data to the pool to free up space in global memory.
3. Metadata is updated in global memory.
4. The VMAX 3 front-end ports process the host write request as a Fast Write.
5. The new writes (metadata) are updated in global memory.
6. VMAX3 sends the status to the host.
7. The VMAX3 back-end ports asynchronously de-stage the data from global memory to the pool.

Lesson Summary

During this lesson, the following topics were covered:

- Read operations: Read hit
- Read operations: Read miss
- FlashBoost
- Write operations: Fast Write
- Write operations: Delayed Fast Write

This lesson covered the read operations – read hit and read miss, and FlashBoost. It also covered the write operations – Fast Write and Delayed Fast Write.

Lesson: VMAX3 Data Integrity Features

This lesson covers the following topic:

- VMAX3 data integrity features

This lesson covers VMAX3 data integrity features.

VMAX3 Data Integrity

Integrity of data in cache is preserved by

- Ongoing data checks and corrections
 - Error Checking and Correction (ECC) code
- Proactive monitoring of the hardware where data is cached
 - Physical memory error correction and verification

Error checking

- Byte level error checking
- Block level error checking

EMC²

© Copyright 2015 EMC Corporation. All rights reserved.

VMAX3 Fundamentals

48

In VMAX3, data integrity is preserved through: Cache Integrity, Error Checking, Drive Integrity, and Vaulting

Cache integrity:

VMAX3 systems preserve the integrity of data stored in the cache by conducting ongoing data checks & corrections and by proactively monitoring the hardware, where the data is cached. This is informally known as **disk scrubbing**. Most storage subsystems add Error Checking and Correction, or ECC bytes to each data record field.

Error checking:

VMAX3 systems correct the **single bit errors** and report an error code once they reach a predefined threshold. When a **multi-bit error** occurs, the VMAX3 system fences the physical memory segment and retrieves the data from mirrored memory. In the event where physical memory replacement is required, the VMAX3 system notifies EMC support and a non-disruptive replacement is performed.

VMAX3 Data Integrity (Cont.)

Drive Integrity

- Proactive scans of data to protect against drive read and write errors
- Drive status monitoring and notification

Vaulting

- Vault operations protect data during a manual power down or unexpected power outage
- On power loss, the VMAX3 system uses the SPS power to write cached data in physical memory onto flash IO modules
- Vaulted images are fully redundant

EMC²

© Copyright 2015 EMC Corporation. All rights reserved.

VMAX3 Fundamentals

49

Drive integrity:

VMAX3 systems proactively scan data to protect against drive read and write errors.

VMAX3 systems validate data at transfer, and also use idle time to read data and to monitor drives. During data and drive monitoring, the director checks the data correction bits for validity.

If a drive read error occurs, the director reads all data on that track to VMAX3 system's physical memory and tests the drive track for errors. The director rewrites the data from physical memory back to the drive. The system maps any bad block(s) detected during the test, thereby avoiding defects in the media.

Vaulting:

A vault operation is triggered when the VMAX3 system is powered down, transitioned offline, or when environmental conditions initiate a vault situation.

Lesson Summary

During this lesson, the following topic was covered:

- VMAX3 data integrity features

This lesson covered VMAX3 data integrity features.

Lesson: Storage Efficiency Features

This lesson covers the following topics:

- Virtual provisioning
- Fully Automated Storage Tiering (FAST)
- Service Level Objectives (SLOs)
- FAST.X
- FAST Array Advisor

© Copyright 2015 EMC Corporation. All rights reserved.

VMAX3 Fundamentals

EMC²

51

This lesson covers VMAX3 storage efficiency features like virtual provisioning, Fully Automated Storage Tiering (FAST), Service Level Objectives (SLOs), FAST.X, and FAST Array Advisor.

Virtual Provisioning

VMAX3 arrays are 100% virtually provisioned

Enables creating and presenting a LUN with more capacity than physically allocated

LUNs created are called thin LUNs



EMC²

© Copyright 2015 EMC Corporation. All rights reserved.

VMAX3 Fundamentals

52

VMAX3 arrays are 100% virtually provisioned. Virtual provisioning enables creating and presenting a LUN with more capacity than is physically allocated to it on the storage array. The LUN created using virtual provisioning is called a **thin LUN** to distinguish it from the traditional LUN which is called a **thick LUN**.

VMAX3 – FAST

Automates identification of active and inactive application data for reallocation

Monitors workloads to identify busy and less-busy data

Promotion/demotion activity is based on achieving Service Level Objectives



EMC²

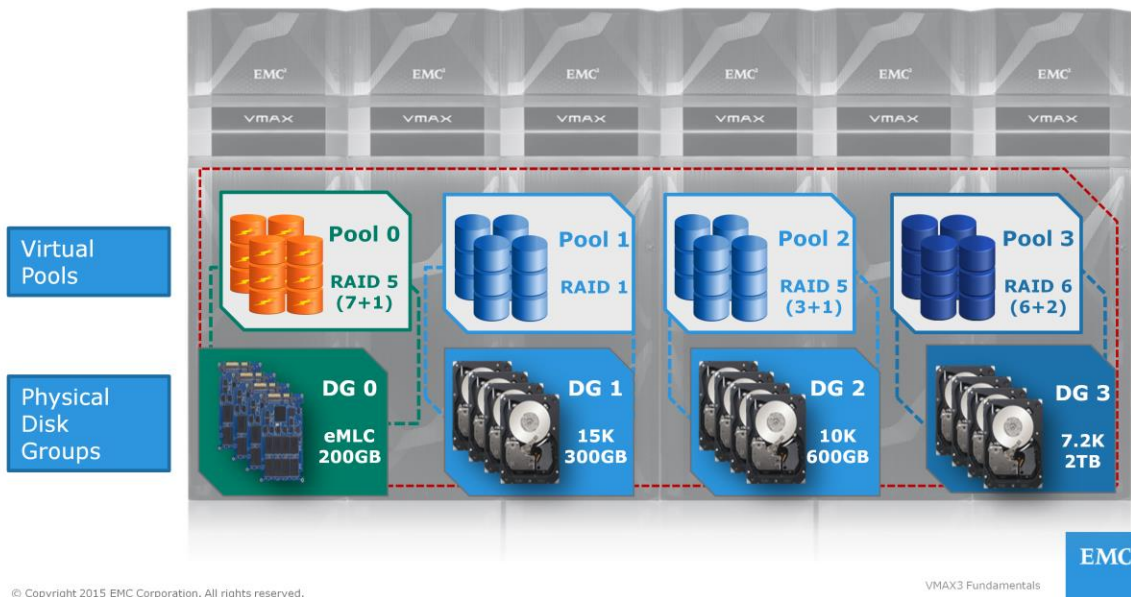
© Copyright 2015 EMC Corporation. All rights reserved.

VMAX3 Fundamentals

53

Fully Automated Storage Tiering (FAST) automates the identification of active or inactive application data for the purpose of reallocating it across different performance/capacity pools within a VMAX3 storage array. FAST proactively monitors workloads to identify most active data that would benefit from being moved to higher performing drives, while also identifying least active data that could be moved to higher-capacity drives, without affecting existing performance. This promotion/demotion activity is based on achieving Service Level Objectives that set performance targets for associated applications, with FAST determining the most appropriate drive technologies or RAID protection types to allocate data on.

VMAX3 – FAST Elements



The main elements of FAST are:

- **Disk groups** - collection of physical drives defined within the VMAX
- **Virtual pools** - logical view of the TDATs

The disk groups and virtual pools are factory preconfigured.

Each drive in a disk group shares the same performance characteristics determined by:

- Rotational speed of hard disk drives (15K, 10K, 7.2K)
- Enterprise Flash Drives (EFD)
- Capacity

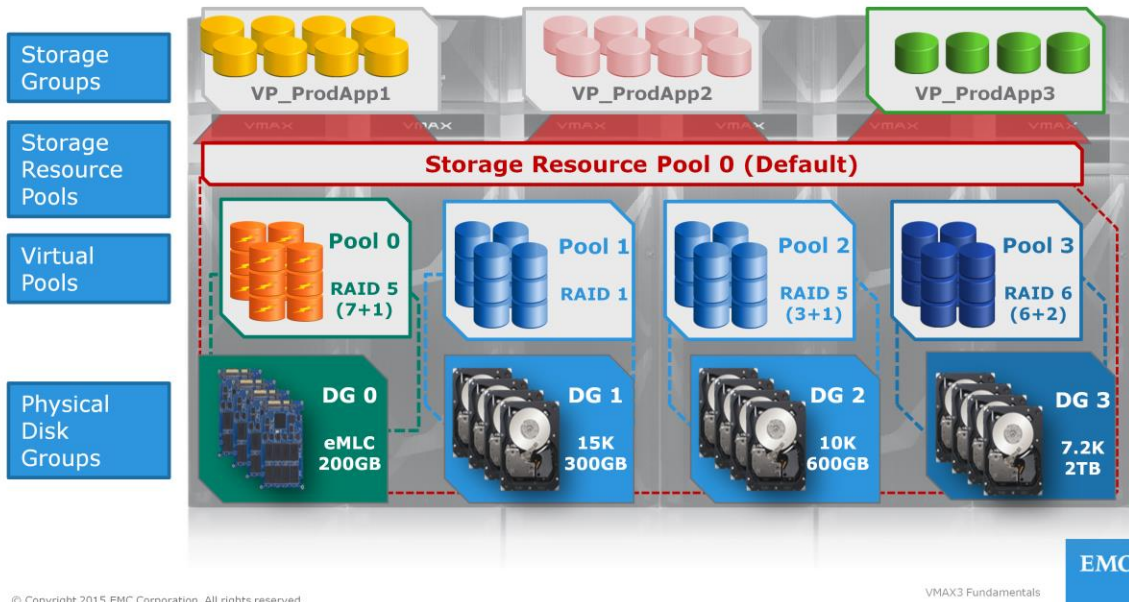
Each disk group is preconfigured with Data Devices (TDATs) and has:

- One single RAID protection
- Fixed size hypervolumes

All TDATs in a disk group are added to a virtual pool. There is a one-to-one relationship between a virtual pool and disk group. The performance capability of each virtual pool is known and is modeled based on the drive type, speed, capacity, quantity of drives, and RAID protection.

Note: Disk groups cannot be modified.

VMAX3 – FAST Elements (cont.)



A Storage Resource Pool (SRP) is a collection of virtual pools that make up a FAST Domain. A virtual pool can only be included in one SRP. SRPs are defined within the bin file and cannot be configured by the customer. The factory preconfigured array includes one SRP that contains all virtual pools in the array. Multiple (current maximum is 2) SRPs may be configured of which one must be marked as the default SRP.

A storage group is a logical grouping of devices for the purpose of common management. They can be used for Masking and/or FAST. A storage group can be associated with an SRP. This allows devices in the storage groups to allocate storage from any pool in the SRP. By default, storage groups are associated with the default SRP.

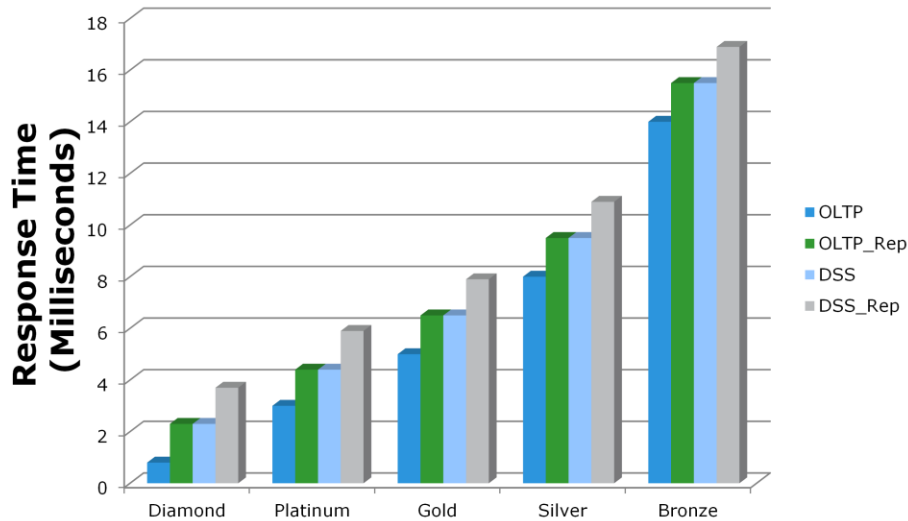
A storage group can be associated with a SLO, which defines the storage group as FAST managed. By default, the storage groups are given the "Optimized" SLO. A thin device may only be in one storage group associated with a SLO and a SRP.

Changing the SRP associated with a storage group will result in all data being migrated to the newly associated SRP. It is possible to configure devices with different SLO policies in the same storage group by the use of sub storage groups. Upon creation, thin devices are associated with the default SRP and given the default SLO.

Devices are in the ready state upon creation. During creation, the device can optionally be added to an existing storage group and the device inherits SRP and SLO from that storage group. No thin device extent is allocated on creation, instead it is only allocated upon first write.

Service Level Objectives

Expected Average Response Time



© Copyright 2015 EMC Corporation. All rights reserved.

VMAX3 Fundamentals

EMC²

56

VMAX3 FAST provides the ability to deliver variable performance levels through Service Level Objectives (SLO). Thin devices can be added to storage groups, and storage groups can be assigned a specific service level objective to set performance expectations. The service level objective defines the response time target for the storage group.

There are six available service level objectives, varying in expected average response time targets of less than one millisecond (Diamond SLO) to no explicit response time target (Optimized). These six service level objectives are fixed and while they may be switched by the user to match the performance goal of the application, they may not be modified to ensure consistency across the platform. The available SLOs include:

- **Diamond** which emulates EFD performance with an expected response time of 0.8 ms
- **Platinum** which emulates performance between EFD and 15K RPM drive at 3.0 ms
- **Gold** which emulates 15K RPM performance at 5.0 ms
- **Silver** which emulates 10K RPM performance at 8.0 ms
- **Bronze** which emulates 7.2K RPM performance at 14.0 ms
- **Optimized** (the default) which achieves optimal performance by placing the most active data on higher performing storage and the least active data on the most cost-effective storage

The SLO can be combined with a workload type to further refine the performance objective.

Workload Types

Workload	Description
OLTP	Small block IO workload
OLTP with Replication	Small block IO workload with local and/or remote replication
DSS	Large block IO workload
DSS with Replication	Large block IO workload with local and/or remote replication
None	No pre-definition of workload type – the system will do the best it can
Custom	Build your own custom workload type based on the real profile of an existing storage group workload



© Copyright 2015 EMC Corporation. All rights reserved.

VMAX3 Fundamentals

57

Workload types typically fall into two broad categories – OLTP and DSS. OLTP stands for Online Transaction Processing. An example of OLTP is an ATM at a bank. OLTP is typically small block IO.

DSS stands for Decision Support System and an example of this is a traditional batch reporting process or query on a database. DSS is typically large block IO. Adding replication, either local or remote, adds an expected latency to IOs.

If you are not sure of the workload type, specify None. The system has knowledge of IO types.

Custom is a workload type that allows the user to build his own custom workload type based on the real profile of an existing storage group workload.

FAST.X Extends VMAX3's Data Services



© Copyright 2015 EMC Corporation. All rights reserved.

VMAX3 Fundamentals

58

VMAX3's data services extend beyond the array and across the data center through **FAST.X**. FAST.X evolves and extends storage tiering and service level management to the storage platforms and the cloud. It allows data movement across storage technologies provided by various block devices that include XtremIO, CloudArray, VNX, or non-EMC storage. It simplifies management and operations, and consolidates heterogeneous storage under its control. It provides the ability to use VMAX features such as zero data loss with SRDF, ProtectPoint, TimeFinder, and VAAI on other arrays.

FAST.X ensures workload optimization without complicated management. It simplifies management at scale and provides workload optimization with the same simplicity as the internal SLO provisioning of VMAX. Similar to advanced automation of FAST which optimizes customer workloads to automatically apply to necessary amount of resources, FAST.X extends this capability to external arrays according to the same specified performance and availability criteria. This simplicity helps to improve overall staff productivity and allows for greater focus on business, rather than managing a multitude of different technologies and storage arrays.

Tiering to EMC XtremIO allows customers to benefit from data reduction of XtremIO (industry-leading all-flash array) while leveraging VMAX3 trust and simplicity. FAST.X to EMC CloudArray enables customers to tier slower or inactive workloads to highly scalable cloud storage for increased cost efficiency and a reduced overall TCO.

FAST.X Integration with XtremIO



© Copyright 2015 EMC Corporation. All rights reserved.

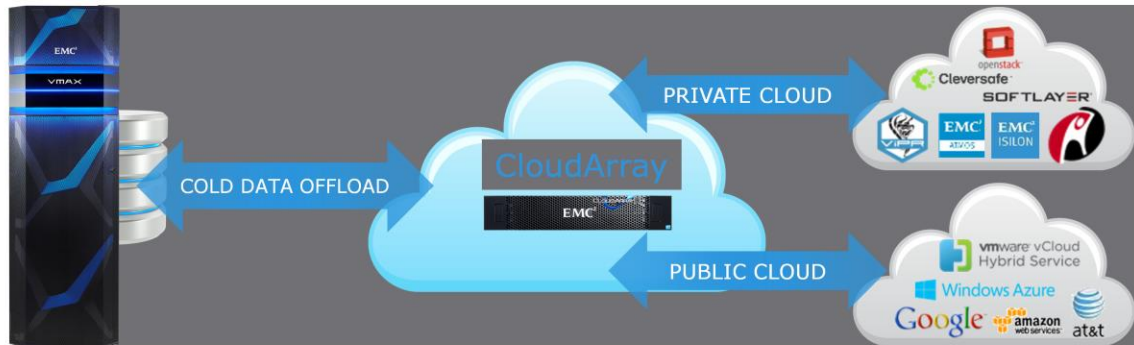
VMAX3 Fundamentals

59

FAST.X with XtremIO is a use case for leveraging external storage for extending data services. Integration of FAST.X with XtremIO on VMAX3 provides a very high performance tier of storage and also gives very effective data reduction. XtremIO is an all-flash array. When XtremIO is configured to use the same storage resource pool (SRP) as the internal storage, the XtremIO storage becomes an extension of the SRP for the flash tier.

VMAX3 FAST.X integration with EMC XtremIO allows customers to benefit from the strengths of both platforms to deliver performance equal to or better than VMAX3 and XtremIO for most workloads. FAST.X with XtremIO also delivers the trusted VMAX protection features.

FAST.X Tiering with CloudArray



Archive inactive data
off VMAX3

Local disk caching
maintains accessibility on
recent data

Move and store data
in a cloud of your
choice

© Copyright 2015 EMC Corporation. All rights reserved.

VMAX3 Fundamentals

EMC²

60

FAST.X with CloudArray enables VMAX3 to provide a cost effective integrated cold data offload capability. In FAST.X with CloudArray integration, the CloudArray must reside in its own storage resource pool (SRP). It is not available as part of the FAST tiering solution, however, CloudArray can be used to offload data which is not frequently used, to the cloud.

FAST.X tiering with EMC CloudArray moves less active workloads to more cost efficient cloud storage, resulting in 40% lesser storage costs with highly scalable backend capacity. This allows VMAX3 to archive inactive data inexpensively to EMC CloudArray and yet, makes the cloud storage look like local storage due to the local caching of recent data on Cloud Array.

The FAST.X with CloudArray integration allows tiering to cloud or object storage in the form of a public or private cloud. The cloud enabled VMAX3 storage lowers the total cost of ownership by moving less frequently used data to the lower cost cloud storage.

Designed to combine the resource efficiency of the cloud with traditional, on premise storage, CloudArray enables organizations to scale their enterprise storage capacity without increasing physical footprint. With support for different public and private cloud platforms, moving and storing data in and out of the clouds becomes seamless.

VMAX3 – FAST Array Advisor



© Copyright 2015 EMC Corporation. All rights reserved.

VMAX3 Fundamentals

EMC²

61

The **FAST Array Advisor** monitors the workload of several VMAX3 arrays. The analysis is combined with the overall performance capability of each array and the contribution of each application to the workloads. The combined analysis is used for making recommendations on applications that could be moved to a less utilized array, bringing a balance to the group of arrays.

Prior to making the recommended relocations, the effect can be modeled with the use of Workload Planner. If the recommendation is accepted, the migration is a manual process that has to be initiated by the administrator.

Lesson Summary

During this lesson, the following topics were covered:

- Virtual provisioning
- Fully Automated Storage Tiering (FAST)
- Service Level Objectives (SLOs)
- FAST.X
- FAST Array Advisor

This lesson covered VMAX3 storage efficiency features like virtual provisioning, Fully Automated Storage Tiering (FAST), Service Level Objectives (SLOs), FAST.X, and FAST Array Advisor.

Lesson: Replication Features

This lesson covers the following topics:

- Local Replication features
 - TimeFinder solutions
 - TimeFinder SnapVX
- Remote Replication features
 - SRDF solutions
 - Enhanced SRDF
 - Enhanced SRDF/A
- VMAX3 – ProtectPoint

This lesson covers the local replication features such as the TimeFinder solutions and TimeFinder SnapVX, and remote replication features such as Symmetrix Remote Data Facility (SRDF), Enhanced SRDF, and Enhanced SRDF/A and VMAX3 – ProtectPoint.

Local Replication – TimeFinder Solutions



- Trusted local replication by industry-leading applications
- Most breadth and depth for array-based local replication



© Copyright 2015 EMC Corporation. All rights reserved.

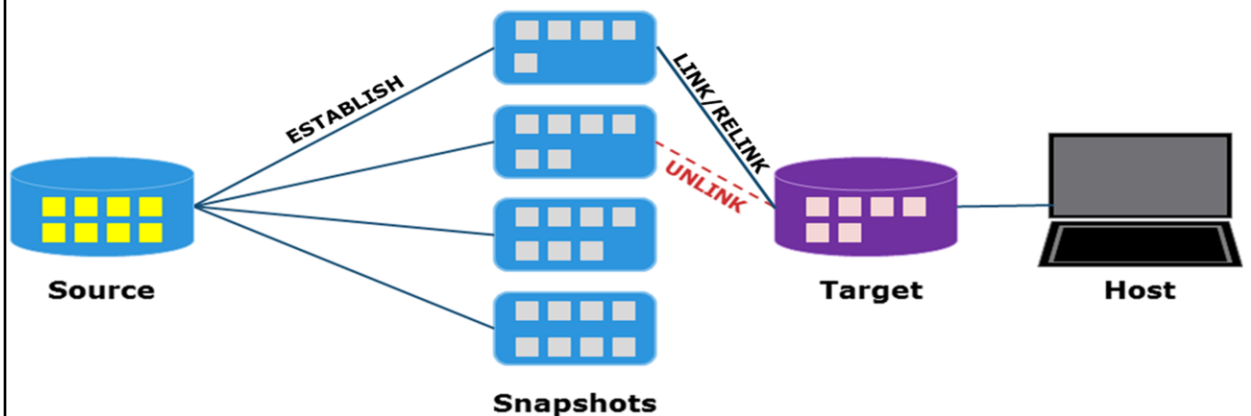
VMAX3 Fundamentals

64

The TimeFinder family has several different solutions to help in meeting service-level requirements, while providing very powerful local replication capabilities. TimeFinder easily integrates into industry-leading applications such as, Oracle, Microsoft, VMware, SAP, and IBM. This is a result of the integration efforts, as well as EMC's partnership with these major application vendors.

TimeFinder is highly recommended with remote replication solutions such as SRDF, to increase application availability, improve test/development capabilities and disaster restart requirements. TimeFinder is the solution VMAX3 customers rely on for business continuity locally, in the array.

Local Replication – TimeFinder SnapVX



© Copyright 2015 EMC Corporation. All rights reserved.

VMAX3 Fundamentals

EMC²

65

TimeFinder SnapVX, new with VMAX3, is a local replication solution designed to non-disruptively create point-in-time copies (snapshots) of critical data. TimeFinder SnapVX creates snapshots by storing changed tracks (deltas) directly in the SRP of the source device. With TimeFinder SnapVX, there is no need to specify a target device and source/target pairs when creating a snapshot. If there is ever a need for the application to use the point-in-time data, links are created from the snapshot to one or more target devices. If there are multiple snapshots and the application needs to find a particular point-in-time copy for host access, link and relink are performed until the correct snapshot is located. Snap operations are controlled from the host by using the Symsnapvx command to create, activate, terminate, and restore the snap copy sessions.

Local Replication – TimeFinder SnapVX

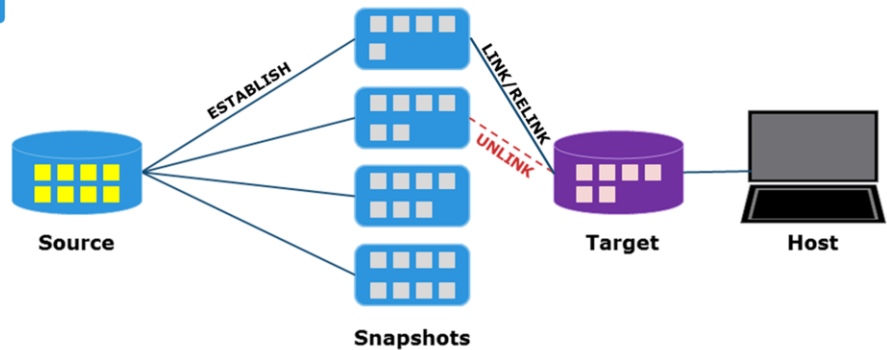
Symsnapvx Operations

Establish

Restore

Linking

Terminate



EMC²

VMAX3 Fundamentals

66

The snap operations described here explain how to manage the devices participating in a copy session using the SYMCLI.

The Symsnapvx command creates point-in-time copies (snapshots) directly in the SRP of the source device. The Symsnapvx operations are establish, restore, link, relink, unlink, and terminate.

- **Establish:** This command creates and activates a TimeFinder SnapVX snapshot with the name that you supply.
- **Restore:** It allows you to restore the snapshot's point-in-time data back to the original source device.
- **Link:** It is used to access a point-in time-copy, a link must be created from the snapshot data to a host mapped target device.
- **Terminate:** Terminating a snapshot removes the snapshot from the system. A snapshot cannot be terminated if it has any links.

Local Replication – TimeFinder SnapVX (cont.)

Legacy modes and commands are supported

Target-less Snapshots (pointer based copies)

Symsnapvx 'Establish' can snap an entire storage group in a single command

Up to 256 snapshots per source

Up to 1024 linked targets per source

Compatibility mode for legacy TimeFinder modes

Snapshot naming and expirations

Copy and no-copy modes

© Copyright 2015 EMC Corporation. All rights reserved.

VMAX3 Fundamentals

EMC²

67

TimeFinder SnapVX utilizes point-in-time copies of volumes that are used for backups, data warehouse (DW) refreshes, DSS, or any other application that needs access in parallel to the production data. It is possible to scale higher since point-in-time pointer based copies are utilized instead of full target volumes. There can be up to 256 snapshots per source volume.

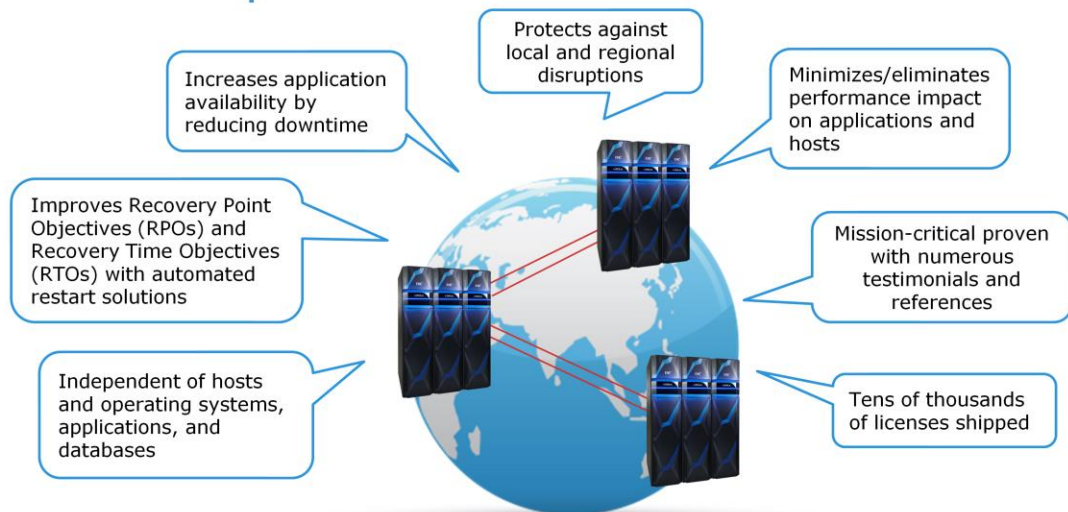
The snapshots may be named to help in identification and an expiration date can be set. SnapVX snapshots cannot be directly accessed by a host. They can be either restored to the source devices or linked to up to 1024 sets of target devices. When linking any snapshot to target devices, SnapVX allows using **copy** or **no-copy** option where no-copy option is the default. Without having to create target devices, the process is simpler and more efficient. Because there are no target devices, the creation and termination times are faster.

Note: The **Write Pacing** feature is no longer supported or needed with TimeFinder SnapVX.

SnapVX supports TimeFinder Clone, VP Snap, and Mirror via emulations that transparently convert commands to SnapVX commands. Scripts that use TimeFinder Clone, VP Snap, and Mirror commands are supported, but the underlying mechanism will be SnapVX. However, it is recommended to change existing scripts to use SnapVX to take advantage of the new features.

Refer to the technical note "EMC VMAX3 Local Replication – TimeFinder SnapVX and TimeFinder Emulation" for additional information.

Remote Replication – SRDF Solutions



Trusted Remote Replication – Information Availability 24x7

EMC²

© Copyright 2015 EMC Corporation. All rights reserved.

VMAX3 Fundamentals

68

The Symmetrix Remote Data Facility (SRDF) family of software is the gold standard for remote replication in mission critical environments. Built for the high-end VMAX hardware architecture, the SRDF family of solutions is trusted for disaster recovery and business continuity. SRDF enables remote data services to provide 6 nines data availability 24x7 operations with synchronous SRDF.

The SRDF family offers unmatched deployment flexibility and massive scalability to deliver a wide range of distance replication capabilities, which helps in meeting mixed service level requirements with minimal effect on operations. The VMAX3 SRDF replication features offer performance, flexibility, consistency, integration, management, and results.

Remote Replication – Enhanced SRDF

VMAX3 to VMAX3	VMAX3 to VMAX
Target can be larger than source	Target can be larger than source
Compression supported	Meta and Non-meta devices supported
	Thick or thin provisioned devices supported
	Compression supported

© Copyright 2015 EMC Corporation. All rights reserved.

VMAX3 Fundamentals

EMC²

69

With VMAX3 and HYPERMAX OS 5977, SRDF has been enhanced to provide increased performance of devices involved in remote replication, removing the need to configure Meta devices or increase device counts. This allows you to achieve desired performance goals with less complex implementation even in the most demanding environments.

Enhanced SRDF includes SRDF support for VMAX3 to VMAX3, and VMAX3 to legacy VMAX with Enginuity 5876 and patch 67492. When SRDF is used to connect to a legacy VMAX, both Meta and Non-meta devices are supported, and the target device can be larger than the source device. A thin device on a VMAX3 system can have a SRDF relationship with a thin device or a thick device on a legacy VMAX system. Compression is supported, as in Dynamic SRDF.

Remote Replication – Enhanced SRDF/A

SRDF/A Multi-Cycle Mode

- Allows multiple capture cycles on the source side
- Capture cycles are independent of Transmit, Receive, and Apply cycles
- VMAX3 to VMAX3 only

© Copyright 2015 EMC Corporation. All rights reserved.

VMAX3 Fundamentals

EMC²

70

Enhancements have been made to SRDF/A, allowing more than two capture cycles on the source side. **SRDF/A Multi-Cycle Mode (MCM)** removes the creation of a new capture cycle from its dependence on the transmit, receive, and apply cycles.

The benefit of this is to capture controlled amounts of data on the source side. Each capture cycle occurs at regular intervals and does not contain large amounts of data waiting for a cycle to occur. Another benefit is, the data that is sent across the SRDF link is smaller in size and does not overwhelm the target side. The target side still has two cycle switches, the receive and the apply.

In order to take advantage of SRDF/A Multi-Cycle Mode, both the source and target arrays must be VMAX3 systems. Both Single-session and Multi-session consistency are supported with SRDF/A MCM.

VMAX3 – ProtectPoint

- Cost friendly retention of data with Data Domain's deduplication
- Protection from corruption
- Application integration with agents that allow application owners to run their own backups
- Zero impact on application servers
- RPO measured in minutes instead of hours with minimal RTO
- Only changed data is sent across SAN

Eliminate Backup Impact on App	Meet Stringent Protection SLAs	Reduce Cost and Complexity
		
Non-intrusive Data Protection Consistent Application Performance	Faster Backup and Recovery Meet SLAs for Mission Critical Applications	Simple and Efficient No Additional Infrastructure Required

EMC²

© Copyright 2015 EMC Corporation. All rights reserved.

VMAX3 Fundamentals

71

ProtectPoint provides the functionality of backups with the performance of a snapshot. Only changed blocks are sent directly across the SAN and all backups are stored in native formats, resulting in fast backup, recovery, and instant access to protected data for simplified and granular recovery. Application integration with agents allow the application owners to run their own backups and gain consistent backups resulting in simplified and expedited recoveries.

Backups result in moving data directly from the VMAX3 to Data Domain eliminating any application server overhead.

Lesson Summary

During this lesson, the following topics were covered:

- Local Replication features
 - TimeFinder Solutions
 - TimeFinder SnapVX
- Remote Replication features
 - SRDF Solutions
 - Enhanced SRDF
 - Enhanced SRDF/A
- VMAX3 – ProtectPoint

This lesson covered the local replication features such as, the TimeFinder solutions and TimeFinder SnapVX, and remote replication features such as, Symmetrix Remote Data Facility (SRDF), enhanced SRDF, and SRDF/A and VMAX3 – ProtectPoint.

Lesson: Embedded NAS (eNAS)

This lesson covers the following topics:

- eNAS with VMAX3
- eNAS management interface
- Replication using eNAS

This lesson covers eNAS with VMAX3, eNAS Management Interface, and replication using eNAS.

eNAS with VMAX3

Capability to host embedded applications on the VMAX

Unifies file and block data services

Configurable through Unisphere for VMAX



EMC²

© Copyright 2015 EMC Corporation. All rights reserved.

VMAX3 Fundamentals

74

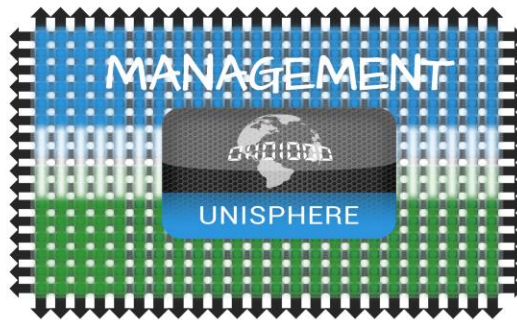
VMAX3 introduces the capability to host embedded applications on the VMAX. It unifies file and block data services via Embedded Network Attached Storage (eNAS) reducing the capital and operational expense, as well as the rack space. The eNAS implementation is fully integrated into the VMAX3 platform and is configurable through Unisphere for VMAX.

The advantages of eNAS are:

- Leverages VMAX3 features (FAST, Host I/O limits)
- Supports pre-installed network-ready software that can be activated after an array is installed
- Provides Unisphere for VMAX streamlined block and file management capabilities, creating a unified user experience that supports:
 - Creating storage pools
 - Managing storage pools
 - Summary view of block and file assets
 - File to block mapping
 - Correlating file and block events and alerts

For detailed information on eNAS, refer to the EMC VMAX3 Unified Embedded NAS technical notes available on <http://www.emc.com>.

eNAS Management Interface



Link and Launch

Provision Storage for File Wizard

© Copyright 2015 EMC Corporation. All rights reserved.

VMAX3 Fundamentals

EMC²

75

eNAS block and file services are managed using Unisphere for VMAX File dashboard. Link and Launch enables users to run block and file management GUI within the same session. The Provision Storage for File wizard helps users create storage groups (automatically provisioned to the Data Movers) quickly and easily. Creating a storage group creates a storage pool in Unisphere for VNX that is used for file level provisioning tasks.

Replication using eNAS

eNAS
replication
with

- VNX Replicator for File
- Virtual Data Mover
-Replication with
SRDF/S
- VNX SnapSure

EMC²

© Copyright 2015 EMC Corporation. All rights reserved.

VMAX3 Fundamentals

76

eNAS replication is performed using:

- VNX Replicator for File – asynchronous file system level replication.
- Virtual Data Mover (VDM) – synchronous replication with SRDF/S and optional automatic failover manager. File Auto Recovery (FAR) with optional File Auto Recover Manager (FARM).
- VNX SnapSure – Create and manage checkpoints (point-in-time, logical images of a production file system).

eNAS replication is available as part of the Remote Replication Suite and Local Replication Suite.

Lesson Summary

During this lesson, the following topics were covered:

- eNAS with VMAX3
- eNAS management interface
- Replication using eNAS

This lesson covered eNAS with VMAX3, eNAS management interface and replication using eNAS.

Lesson: Security Features

This lesson covers the following topics:

- Security overview
- Secure audit
- User authorization
- Data at Rest Encryption (D@RE)
- Data Erasure

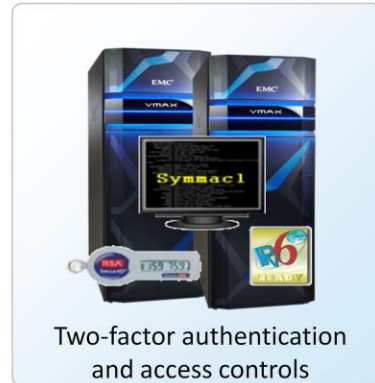
This lesson covers security features that the VMAX3 family offers, which includes secure audit, user authorization, Data at Rest Encryption (D@RE), and Data Erasure.

Security Overview

Secure Audit



Secure Access



© Copyright 2015 EMC Corporation. All rights reserved.

VMAX3 Fundamentals

EMC²

79

VMAX3 family security features support the latest security protocols and also provide robust authentication, audit logging, and data erasure services.

For secure audit, EMC has tamper-proof logs and RSA enVision integration.

For secure access, EMC has two-factor authentication and access controls.

VMAX3 Service Credential secured by RSA is an example of EMC delivering integrated technology with its RSA Security portfolio. RSA features prevent unauthorized service actions on VMAX3 systems through integrated RSA technology. It provides authentication of user identities and authorizes the actions of EMC and partner service personnel as they log into the service processor.

Secure Audit

Audit Log Records

- Host-initiated actions
- Physical component changes
- Actions on the MMCS
- D@RE key management events
- Attempts blocked by security controls (Symmetrix Access Controls)

Secure Audit



© Copyright 2015 EMC Corporation. All rights reserved.

VMAX3 Fundamentals

80

For secure audit, EMC has tamper-proof logs and RSA EnVision integration. The audit log records major activities on the VMAX3 array, including:

- Host-initiated actions
- Physical component changes
- Actions on the MMCS
- Data At Rest Encryption (D@RE) key management events
- Attempts blocked by security controls (Symmetrix Access Controls)

Event contents cannot be altered. Users with the auditor access (assigned using Symmetrix Access Controls) can view but not modify the log.

User Authorization

Users based on usernames are mapped to specific roles that define permitted operations:

Roles
1. None
2. Storage Admin
3. Administrator
4. Auditor
5. Performance Monitor
6. Monitor
7. Security Admin

User-based access control

Roles and usernames are used by both Solutions Enabler CLI and Unisphere for VMAX

Configured with Solutions Enabler CLI or Unisphere for VMAX

Configured independently for each VMAX3 array

EMC²

© Copyright 2015 EMC Corporation. All rights reserved.

VMAX3 Fundamentals

81

An alternative to host-based access control provided by VMAX3 access control is user-based authorization. With user-based authorization each role is associated with a username. Based on the role, the user has a specified access level for all devices in VMAX. Host-based and user-based access control, while completely separate and independent, can be used together to create the highest level of security.

Data At Rest Encryption (D@RE)



- Encrypts all block/file data on the array at the drive level
- Supports all VMAX3 data services and drive types
- Uses RSA embedded encryption key manager

Protects
against
DRIVE LOSS

- ✓ Federal mandate
- ✓ Industry compliance
- ✓ Often eliminates drive shredding

EMC²

© Copyright 2015 EMC Corporation. All rights reserved.

VMAX3 Fundamentals

82

Data at Rest Encryption (D@RE) provides hardware-based, on-array back-end encryption to protect data from unauthorized access. Controller-based D@RE encrypts the entire array with minimal performance overhead and works with both file and block data. D@RE uses SAS I/O modules with Advanced Encryption Standard (AES) – 256-bit encryption designed to be FIPS 140-2 Level 1 compliant.

D@RE encrypts and decrypts data as it is being written to or read from the disk. When D@RE is enabled, all configured drives are encrypted including data drives, spares, and drives with no provisioned volumes.

D@RE enables:

- **Secure replacement of failed drives that cannot be erased**
For some types of hard drive failures, data erasure is not possible. Without D@RE, if the failed drive is repaired, data on the drive may be at risk. With D@RE, simply delete the applicable keys and the data on the failed drive is unreadable.
- **Protection against stolen drives**
When a drive is removed from the array, the key stays behind making data on the drive unreadable.
- **Faster drive sparing**
The drive replacement script destroys the keys associated with the removed drive, making data on that drive unreadable.
- **Secure array retirement**
Simply delete all copies of keys on the array and all remaining data is unreadable.

D@RE is compatible with all VMAX3 array features and all supported local drive types or volume emulations. Encryption is a powerful tool for enforcing security policies. D@RE delivers encryption without degrading performance or disrupting existing applications and infrastructure.

Data Erasure

Mitigates the risk of information dissemination and helps secure information at the end of the information lifecycle

Protects data from unauthorized access

Ensures secure data migration by making data on the source array unreadable

Supports compliance with internal policies and regulatory requirements

EMC²

© Copyright 2015 EMC Corporation. All rights reserved.

VMAX3 Fundamentals

83

EMC Data Erasure uses specialized software to erase information on VMAX3 arrays. This service delivers certified erasure of disks before they are proactively removed from the Symmetrix and replaced. As drives fail within the Symmetrix, specialized software automatically relocates the data to a spare drive then erases the information from the original disk before there is an opportunity for a breach. Data erasure mitigates the risk of information dissemination and helps secure information at the end of the information lifecycle.

Data erasure:

- Protects data from unauthorized access
- Ensures secure data migration by making data on the source array unreadable
- Supports compliance with internal policies and regulatory requirements

Data Erasure overwrites data at the lowest application-addressable level to disk drives. The number of overwrites is configurable from 3x (the default) to 7x with a combination of random patterns on the selected arrays. Overwrite is supported on Fibre Channel, SAS, and Flash drives. An optional certification service is available to provide a certificate of erasure. Drives that fail erasure are delivered to customers for final disposition.

Lesson Summary

During this lesson, the following topics were covered:

- Security overview
- Secure audit
- User authorization
- Data at Rest Encryption (D@RE)
- Data Erasure

This lesson covered the security features the VMAX3 family offers which includes secure audit, user authorization, Data at Rest Encryption and Data Erasure.

Module Summary

Key points covered in this module:

- Describe VMAX3 key features and capabilities
- Identify the usage of key features and capabilities in an IT environment

This module covered VMAX3 functions and features such as read/write operations, data integrity, storage efficiency, local & remote replication, integration of eNAS with VMAX3, and security.

Module: VMAX3 Management Overview

Upon completion of this module, you should be able to:

- Explain the various Software Suites
- Describe VMAX3 management options
- Explain the various VMAX3 external management tools

This module focuses on various VMAX3 Software Suites, management options and external management tools.

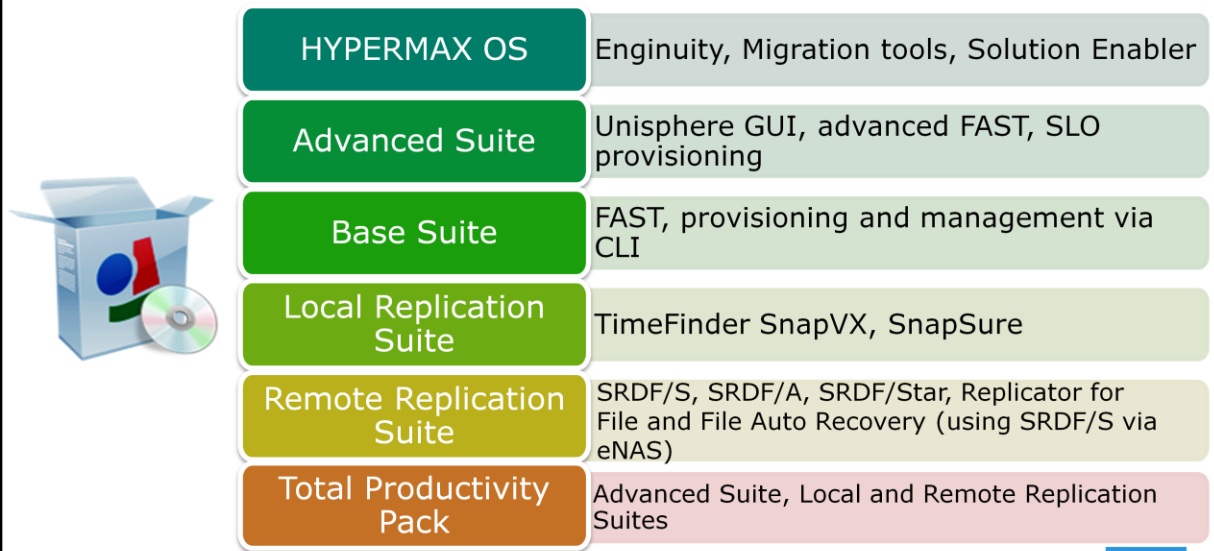
Lesson: Management Options

This lesson covers the following topics:

- VMAX3: Software suites
- Solutions Enabler
- Unisphere for VMAX
- Database Storage Analyzer
- SMI-S Provider
- eManagement (Embedded Unisphere)

This lesson covers VMAX3 Software suites, Unisphere for VMAX, Solutions Enabler, SMI-S Provider, eManagement (Embedded Unisphere) and Database Storage Analyzer.

VMAX3 Software Suites



© Copyright 2015 EMC Corporation. All rights reserved.

VMAX3 Fundamentals

EMC²

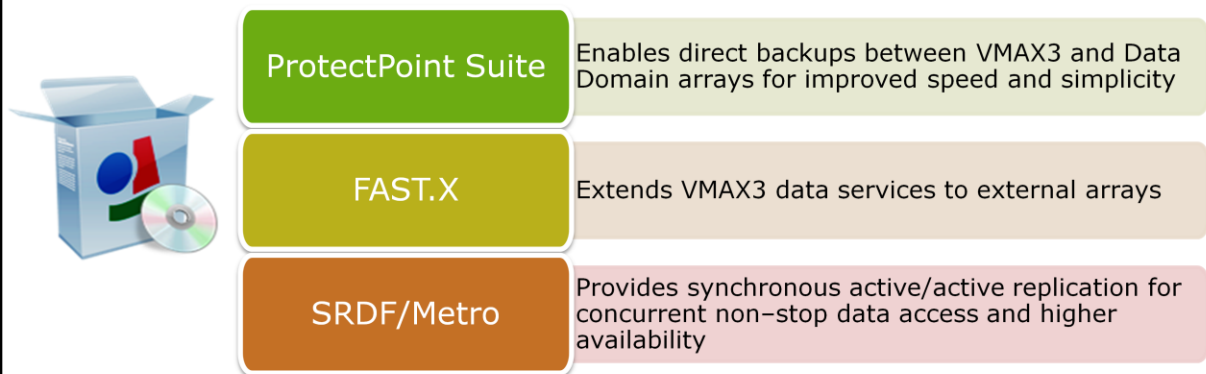
88

The VMAX3 software packaging comes in a variety of suites. Each suite contains a unique set of solutions to improve efficiency and availability by simplifying and automating many storage tasks. Functionality is purchased in the form of a suite and combinations of suites are packaged into software packs.

The suites available for VMAX3 arrays are:

- HYPERMAX OS
- Advanced Suite
- Base Suite
- Local Replication Suite
- Remote Replication Suite
- Total Productivity Pack which is a package of Advanced Suite, Local and Remote Replication Suites

VMAX3 Additional Software Options



© Copyright 2015 EMC Corporation. All rights reserved.

VMAX3 Fundamentals

EMC²

89

ProtectPoint, FAST.X and SRDF/Metro are separately charged items and not included in any of the software suites or the Total Productivity Pack.

ProtectPoint – By integrating VMAX3 and Data Domain storage, ProtectPoint reduces cost and complexity by eliminating traditional backup servers and applications while still providing the benefits of native backups, faster recovery times, and instant access to backups from Data Domain for simple granular recovery.

FAST.X – VMAX3 extends data services like SRDF, SnapVX, ProtectPoint, eNAS and automated storage tiering to external storage platforms from EMC and third-party vendors.

SRDF/Metro – Built upon SRDF/S, SRDF/Metro provides synchronous active/active replication across metro distances for non-stop data access and higher availability. This provides concurrent access of storage groups and LUNs for seamless data mobility and application failover.

Solutions Enabler

Provides the host with the Symmetric Command Line Interface (SYMCLI), including the SYMAPI shared libraries

It is a pre-requisite for other management software including Unisphere for VMAX

Solutions Enabler installer also includes the EMC VSS Provider and the SMI-S Provider

Intended to manage various types of control operations



EMC²

© Copyright 2015 EMC Corporation. All rights reserved.

VMAX3 Fundamentals

90

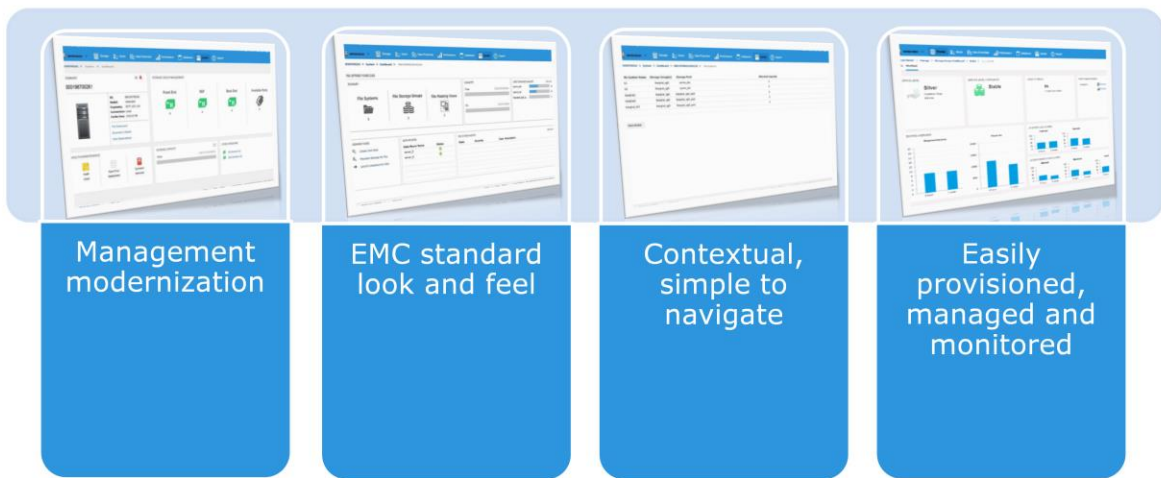
The VMAX3 system provides a management abstraction whereby physical resources are optimized to provide optimal levels of performance, capacity utilization, and seamless mobility.

The **Solutions Enabler kit** is the software that provides the host with the Symmetrix Command Line Interface (SYMCLI), including the SYMAPI shared libraries. SYMCLI is a comprehensive command set for managing EMC enterprise storage environment. SYMCLI supports the VMAX3 and VMAX Family arrays with HYPERMAX OS and Enginuity.

Solutions Enabler is a pre-requisite for other management software including Unisphere for VMAX.

Solutions Enabler installer also includes the EMC VSS Provider and the SMI-S Provider. Solutions Enabler is intended for use by advanced command-line users and script programmers to manage various types of control operations on VMAX3 arrays and devices using the SYMCLI commands of the Solutions Enabler software.

Unisphere for VMAX



© Copyright 2015 EMC Corporation. All rights reserved.

VMAX3 Fundamentals

EMC²

91

Unisphere for VMAX is EMC's management console for the VMAX3 Family. It offers navigation and streamlined operations to simplify and reduce the time required to manage your data center. It establishes EMC's direction to simplify management under a common framework. The VMAX3 Management Suite enables customers to use Unisphere for VMAX using a single eLicensing entitlement.

With Unisphere, VMAX3 customers can provision, manage, monitor, and analyze VMAX3 arrays from one console, significantly reducing storage administration time. Navigation through the product is easier, faster, and more intuitive, and the learning curve is greatly reduced for those users who also work with other Unisphere management consoles.

Unisphere for VMAX supports base array management functionality, such as auto-provisioning, virtual provisioning, FAST controls and performance monitoring. Analysis and performance monitoring capabilities are also included. It supports base array management functionality, such as auto-provisioning, virtual provisioning, FAST controls and performance monitoring.

Provides the ability to monitor and manage SRDF and TimeFinder replication operations.

Database Storage Analyzer

Database Storage Analyzer (DSA) is an application that provides a database to storage performance troubleshooting solution for Oracle and SQL Server databases running on EMC Symmetrix and VMAX storage systems

DSA is bundled with Unisphere for VMAX



EMC²

© Copyright 2015 EMC Corporation. All rights reserved.

VMAX3 Fundamentals

92

Database Storage Analyzer (DSA) is an application that provides a database to storage performance troubleshooting solution for Oracle and SQL Server databases running on EMC Symmetrix and VMAX storage systems.

It is a feature of the Unisphere for VMAX Foundation Suite. It supports database to storage correlation by providing a shared view of how performance issues correlate to database and storage level activities. This view is accessible by database administrators (DBAs) and storage administrators (SAs). The view presents I/O metrics such as input/output operations per second (IOPS), throughput and response time from both the data base and the storage system, which helps to immediately identify gaps between the database I/O performance and the storage I/O performance.

SMI-S Provider

Facilitates the management of storage devices from multiple vendors in storage area networks (SANs).

Examples of applications which use the EMC SMI-S provider are EMC ViPR SRM and EMC Virtual Storage Integrator(VSI) for VMware vSphere Web Client



EMC²

© Copyright 2015 EMC Corporation. All rights reserved.

VMAX3 Fundamentals

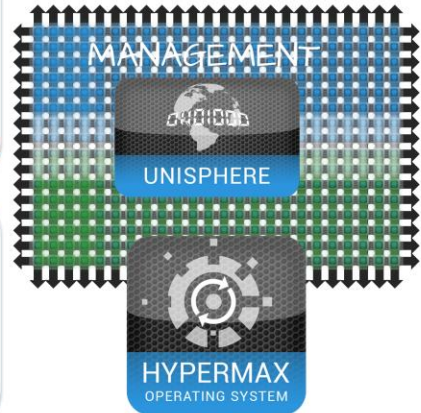
93

SMI-S (Storage Management Initiative Specification) is a standard developed by the Storage Network Industry Association (SNIA) that is intended to facilitate the management of storage devices from multiple vendors in storage area networks (SANs). SMI-S facilitates the management of storage devices from multiple vendors in storage area networks (SANs). SMI-S Provider is a common, standards-based management interface that permits third party applications to configure and manage functions within a storage array. Examples of applications which use the EMC SMI-S provider are EMC ViPR SRM and EMC Virtual Storage Integrator(VSI) for VMware vSphere Web Client.

eManagement (Embedded Unisphere)

This feature enables the customers to further simplify management, reduce cost, and increase availability by running VMAX3 management software directly on VMAX3 arrays

Support Unisphere, Solutions Enabler, vApp and SMI-S without the requirement of a separate server



© Copyright 2015 EMC Corporation. All rights reserved.

VMAX3 Fundamentals

EMC²

94

eManagement (Embedded Unisphere) enables the customers to further simplify management, reduce cost, and increase availability by running VMAX3 management software directly on VMAX3 arrays.

Lesson Summary

During this lesson, the following topics were covered:

- VMAX3: software suites
- Solutions Enabler
- Unisphere for VMAX
- Database Storage Analyzer
- SMI-S Provider
- eManagement (Embedded Unisphere)

This lesson covered VMAx3 Software suites, Unisphere for VMAX, Solutions Enabler, SMI-S provider, eManagement (Embedded Unisphere) and Database Storage Analyzer.

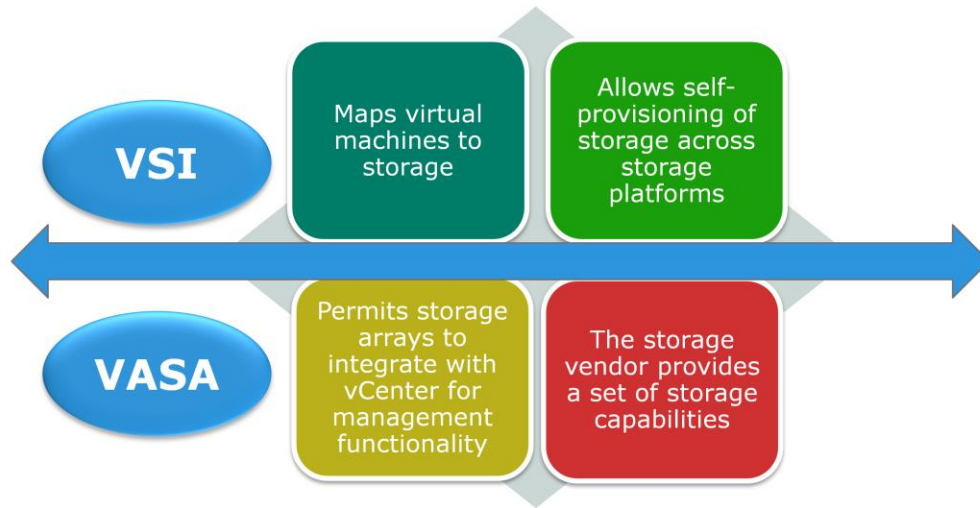
Lesson: External Management Tools

This lesson covers the following topics:

- Integration with VSI & VASA
- SRDF Adapter for VMware vCenter Site Recovery Manager
- ViPR SRM and VMAX
- ViPR Controller and VMAX

This lesson covers integration with VSI & VASA, SRDF adapter for VMware vCenter Site Recovery Manager ViPR SRM and Controller for VMAX.

VSI & VASA



EMC²

VMAX3 Fundamentals

97

Virtual storage integrator (VSI) is a no-charge VMware vCenter plug-in available to all VMware users with EMC storage in their environments. VSI enables IT organizations to achieve simplicity and efficiency in data center operations. VSI dramatically simplifies management of virtualized storage with the ability to map virtual machines to storage and to self-provision storage across EMC Symmetrix VMAX, EMC VNX, and EMC XtremIO families of storage platforms from VMware vCenter.

VASA (vSphere storage API for Storage Awareness) is a set of VMware APIs that permits storage arrays to integrate with vCenter for management functionality. When you register a VASA Provider in vCenter, the storage vendor provides a set of storage capabilities that it uses to classify the devices presented to the VMware environment. EMC implements VASA through the SMI-S Provider. Once installed, the vApp or the host environment requires access to gatekeepers which are small devices presented from the Symmetrix array(s) used in the VMware environment. The gatekeepers allow communication with the array.

SRDF Adapter for VMware vCenter Site Recovery Manager



© Copyright 2015 EMC Corporation. All rights reserved.

VMAX3 Fundamentals

EMC²

98

SRDF Adapter is a Storage Replication Adapter (SRA) that extends the disaster restart management functionality of VMware vCenter Site Recovery Manager (SRM) to the EMC VMAX storage environment. It allows Site Recovery Manager to automate storage-based disaster restart operations on VMAX arrays in an SRDF configuration. SRDF Adapter Utilities enable you to perform many systems-related and storage-related tasks from within vSphere Client, eliminating the need for external tools and management applications.

SRDF Adapter Utilities include the functions listed below:

- **Global Options file editing**

The SRDF Adapter Global Options file can be edited from within the web interface.

- **Automated consistency groups creation**

When using SRDF with Site Recovery Manager, Symmetrix device consistency groups can be automatically created and configured to enable failback and assist with failover preparation and configuration.

- **Failover test configuration**

Device pairs can be defined, which SRDF Adapter uses when testing recovery plans with EMC TimeFinder technology.

- **Device masking and unmasking control**

An SRDF adapted configuration file can be generated to control masking and unmasking of VMAX LUNs.

ViPR SRM and VMAX



ViPR SRM provides end-to-end topology view and data path details for VMAX3.

Visibility helps in performance analysis and in identifying the bottlenecks.

© Copyright 2015 EMC Corporation. All rights reserved.

VMAX3 Fundamentals

EMC²

99

ViPR SRM offers a rich end-to-end topology view and data path details used to streamline performance analysis for VMAX. The topology view provides an understanding of the relationship dependencies and path details from the host to the array. It helps in selecting objects along the data path and presents a number of report options for storage administrators.

Depending on the object selected, various reports are generated such as attributes, storage capacity, performance, path details, and storage connectivity. This visibility makes it much easier for analyzing performance along the data path to help administrators identify where the bottlenecks exist. It also eliminates the need for opening every individual element manager to manually consolidate a performance report.

ViPR Controller and VMAX



Transforms heterogeneous storage into simple, extensible and open storage platforms

Enables to deliver storage as a service to distribute the application workload

© Copyright 2015 EMC Corporation. All rights reserved.

VMAX3 Fundamentals

EMC²

100

ViPR Controller is a storage automation software that centralizes and transforms heterogeneous multi-vendor storage into simple, extensible, and open storage platforms. It allows abstraction from the storage platform underneath it. This enables to deliver storage as a service to distribute the application workload. ViPR does this by defining a series of storage arrays across the infrastructure which might include different storage options like Isilon, VMAX, VNX, and even third party vendor arrays.

The services offered by ViPR Controller are:

- Automated storage provisioning
- Heterogeneous storage
- Single point of management

Lesson Summary

During this lesson, the following topics were covered:

- Integration with VSI & VASA
- SRDF Adapter for VMware vCenter Site Recovery Manager
- ViPR SRM and VMAX
- ViPR Controller and VMAX

This lesson covered integration with VSI & VASA, SRDF adapter for VMware vCenter Site Recovery Manager and ViPR SRM and Controller for VMAX.

Module Summary

Key points covered in this module:

- Explain the various Software Suites
- Describe VMAX3 management options
- Explain the various VMAX3 external management tools

This module covered various VMAX3 Software Suites, management options and external management tools.

Course Summary

Key points covered in this course:

- VMAX3 solution offered by EMC
- Key use cases supported by the VMAX3
- VMAX3 architecture
- Hardware and software options
- Features and hardware/software modules
- Management options and capabilities

© Copyright 2015 EMC Corporation. All rights reserved.

VMAX3 Fundamentals

EMC²

103

This course covered the EMC VMAX3 series. It includes the EMC VMAX3 series models, architecture, features, functions, capabilities, and management.